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Invited Research Article

High-temperature deformation of clinopyroxene–plagioclase aggregates under wet conditions



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Highlights

- The deformation tests use a Paterson-type gas medium apparatus under high temperature and wet conditions.
- The dislocation creep is the primary deformation mechanism, accompanied by dissolution-precipitation creep.

- The data supports the “jelly sandwich” model of the continental lithosphere.

Abstract

Understanding the rheological properties of the continental lower crust (CLC), particularly its response to water, is vital for comprehending various geodynamic processes within the Earth. In this study, two groups of clinopyroxene–plagioclase aggregates, which serve as representative samples of the CLC, were hot pressed in a Paterson gas-medium apparatus with “as is” and additional water. Subsequently, deformation tests were conducted on the hot-pressed samples in the same apparatus, under a confining pressure of 300MPa, temperature of 1173–1423K, and strain rates of 6×10^{-6} to $2 \times 10^{-4} \text{ s}^{-1}$. Mechanical data revealed that axial stresses between 15 and 464MPa and total accumulated finite strain values between 12% and 20%. The resulting mechanical data were analyzed and fitted to a flow law, yielding a stress exponent (n) of 3.6 ± 0.6 and activation energy (Q) of $504 \pm 85 \text{ kJ/mol}$ for the “as is” samples, and $n = 3.7 \pm 1.4$ and $Q = 347 \pm 48 \text{ kJ/mol}$ for water-added samples. Microstructural observations revealed that the deformed samples exhibited a shape-preferred orientation (SPO) and crystallographic preferred orientation (CPO) with subgrain boundaries in coarse-grained domains (constituting $\sim 80 \text{ vol\%}$), indicating that the primary deformation mechanism is dislocation creep. Additionally, the presence of melt, neoformed phases (olivine+orthopyroxene) and abundant fine-grained particles ($< 10 \mu\text{m}$) all show subsidiary dissolution-precipitation creep, and the subsidiary dissolution-precipitation creep played an important role under higher temperature and higher water content conditions. The overall reduction in average grain size resulted in deformation under the grain-size-sensitive creep regime, leading to a notable decrease in sample strength. Overall, these findings confirm the “jelly sandwich” concept of the continental lithosphere by showing that the lower crust is relatively weaker than the upper crust and uppermost mantle.



Keywords

Clinopyroxene–plagioclase aggregates; Water content; Dislocation creep; Rheology; Lower crust

1. Introduction

The continental lower crust (CLC) is an important component of the lithosphere, connecting the felsic upper crust and the ultramafic upper mantle. The CLC's rheological properties have a significant impact on the evolution of the lithosphere ([Christensen and Mooney, 1995](#); [Rudnick and Fountain, 1995](#); [Rudnick and Gao, 2003](#)) and significantly influences various geological processes, such as mechanical coupling between crust and mantle ([Copley et al., 2011](#); [Flesch et al., 2005](#); [Le Breton et al., 2017](#); [Royden, 1996](#)), channel flow ([Getsinger et al., 2013](#); [Shapiro et al., 2004](#)), delamination and underplating ([Magni et al., 2013](#); [Meissner and Mooney, 1998](#); [Wang et al., 2018](#); [Xiao et al., 2019](#); [Zhai et al., 2007](#)) and seismic activity ([Petley-Ragan et al., 2019](#); [Reyners et al., 2007](#); [Shi et al., 2018](#)). Considering that the water content in rock-forming minerals of CLC was found to range from a few hundred ppm to as high as 1900ppm ([Xia et al., 2006](#); [Yang G C et al., 2012](#); [Yang X Z et al., 2008](#); [Zhang et al., 2018, 2016](#)), and CLC's response to water can significantly affect its strength. This has important implications for models of continental rheology ([Bell and Rossman, 1992](#); [Wang et al., 2012](#)), as seismic source depth distributions suggest either a strong lower crust ([Jackson, 2002](#); [Maggi et al., 2000a](#), [Maggi et al., 2000b](#)) or, conversely, a weaker lower crust compared to a strong upper mantle ([Chen and Molnar, 1983](#); [Kohlstedt et al., 1995](#)).

Laboratory-derived flow laws provide valuable insights into the rheological behavior of the CLC ([Bürgmann and Dresen, 2008](#); [Jackson, 2002](#)). Numerous deformation experiments have been conducted on rock-forming minerals of the CLC such as feldspar ([Baïssat et al., 2024](#); [Rybacki and Dresen, 2000](#); [Rybacki et al., 2006, 2010](#)), and pyroxene ([Bystricky and Mackwell, 2001](#); [Chen et al., 2003](#); [Dimanov et al., 2003](#)). These experiments have also focused on olivine, which is the dominant rock-forming mineral in the upper mantle ([Hansen et al., 2012a](#), [Hansen et al., 2012b](#); [Hirth and Kohlstedt, 2013](#); [Karato and Jung, 2003](#); [Korenaga and Karato, 2008](#)).

However, second phases controlled microstructural evolution and flow stress in polymineralic rocks ([Olgaard and Evans, 1986, 1988](#); [Herwegh et al. 1998; 2001, 2004, 2005a,b, 2008, 2011](#); [Linckens et al., 2011](#)), single-phase mineral experiments cannot capture the complex deformation processes observed in multiphase rocks, specifically two-phase aggregates. These aggregates exhibit a diverse mechanical behavior that depends on the content of the second phase and thermodynamic conditions.

Furthermore, their strength is typically restricted within the range of uniform stress and uniform strain rate ([Tullis et al., 1991](#)).

This observation is supported by laboratory findings on anorthite-quartz ([Xiao et al., 2002](#)), calcite-quartz mixtures ([Siddiqi, 1997](#); [Dresen et al., 1998](#); [Rybacki et al., 2003](#)), forsterite-enstatite aggregates ([Ji et al., 2000](#)), and anorthite-diopside aggregates ([Dimanov et al., 2005](#); [2007](#)).

In contrast, limited deformation experiments have been dedicated to investigating granulite, which is representative in CLC ([Ross and Wilks, 1996](#); [Wang et al., 2012](#); [Wen et al., 2021](#); [Wilks and Carter, 1990](#); [Zhou et al., 2017](#)). Among these studies, [Wilks and Carter \(1990\)](#) and [Ross and Wilks \(1996\)](#) performed triaxial compression and shear tests respectively, on an orthopyroxene granulite using a Griggs-type solid-medium apparatus. However, these two studies may not be entirely reliable due to the dominance of a semibrittle deformation mechanism in coarse-grained natural rocks and the inadequate quantification of water content ([Wang et al., 2012](#)). Therefore, Wang et al. (2012) conducted systematic deformation experiments on synthetic mafic granulite using the Griggs solid-medium apparatus and characterized the rheological behavior of mafic granulite during dislocation creep under wet conditions. However, the mafic granulite found in this study is shown to be weaker than the end member of feldspar and pyroxene, indicating a significantly weaker lower crust. Deformation experiments were conducted on natural gabbro ([Zhou et al., 2012](#)) and granulite ([Dang et al., 2024](#); [Zhou et al., 2012](#)) using a gas-medium machine. These experiments revealed that the strength of the granulite is much higher, by comparing the strength of semibrittle deformed granulite ([Ross and Wilks, 1995, 1996](#); [Wilks and Carter, 1990](#)) and the hot-pressed granulite ([Wang et al., 2012](#)) with a comparable composition, smaller grain size, and higher water content.

More recently, [Wen et al. \(2021\)](#) used a Griggs-type solid-medium apparatus to conduct deformation experiments on hot-pressed felsic granulite, and their findings indicate that the strength of felsic granulite is similar to that of mafic granulite as determined by [Wang et al. \(2012\)](#). The inconsistency observed in these previous studies may be attributed to variations in water content, sample composition, grain size, and different deformation apparatus, thereby limiting our understanding of the rheology of CLC.

In order to quantify the effect of water on the rheology of the granulite, our initial step involved conducting hot-pressing experiments on two groups of granulite samples. These experiments were performed using a Paterson gas-medium apparatus at the Guangzhou Institute of Geochemistry, Chinese Academy of Science. The samples were divided into two groups: one with the original water content and the other with additional water. To ensure accurate and reliable data, we carefully selected plagioclase and clinopyroxene with a narrow grain-size range for the hot-pressing process. Subsequently, deformation experiments were conducted on the hot-pressed samples using the same apparatus, allowing us to obtain precise mechanical data.

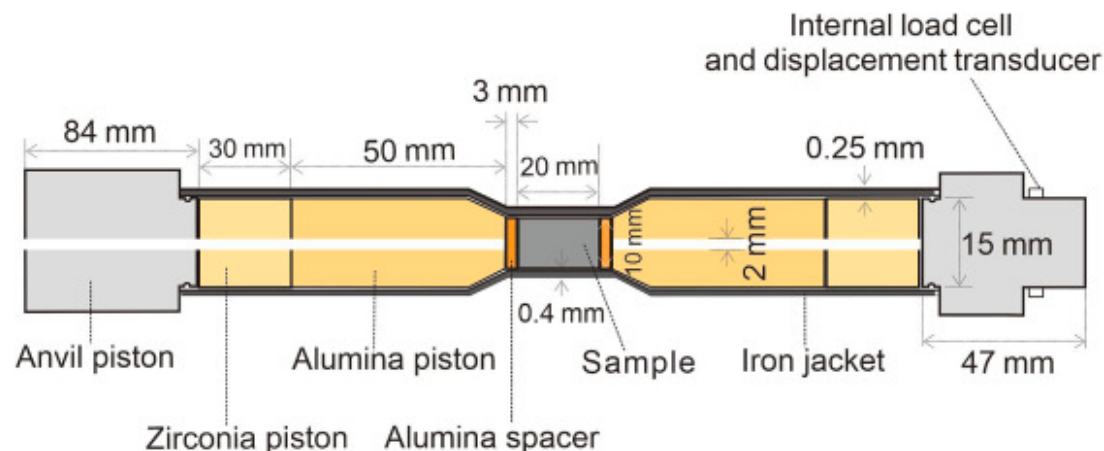
2. Experimental samples and methods

2.1. Starting material

The specimens were derived from a natural granulite (52%Pl+40%Py +3%Q +5%Mag) utilised by [Zhou et al. \(2017\)](#) and obtained from Wayaokou village, Huai'an, Hebei province, China (GPS location: 40°58'20.68"N,114°31'57.72"E). The sample was crushed into powders, with quartz and magnetic separation carried out on the original material. Plagioclase and clinopyroxene were separated using a heavy liquid and ground into fine powders using an agate mortar. After that, these powders were separated using 200-mesh and 250-mesh screens to get the desired grain size of 64–80µm and washed with alcohol multiple times to eliminate any impurities such as dust. Subsequently, we mixed the separated clinopyroxene and plagioclase particles in proportions resembling natural granulite to a ratio of 56:44. Finally, the sorted powder was dried in a forced air oven at 423K for a minimum of 15h for future use.

2.2. Sample preparation and assembly

Samples were prepared with two groups based on water contents: “as is” samples and water-added samples. All powders were cold-pressed into a cylindrical nickel can of 20-mm long and 10-mm diameter. In the case of the water-added samples, 1 wt% of distilled H₂O was added to the powder during the compression process. Each cylindrical sample was placed between two 3-mm-thick alumina spacers and fitted with two 58-mm long alumina pistons and two 30-mm long zircon pistons ([Fig. 1](#)). Subsequently, the complete assembly was encased in a low-carbon iron jacket in order to separate the specimen from the pressure medium. The jacket had a wall thickness measuring 0.25 mm and an inner diameter of 15 mm. Afterwards, the cold-pressed samples underwent a 4-h treatment above a temperature of 1423K and a confining pressure at 300MPa.



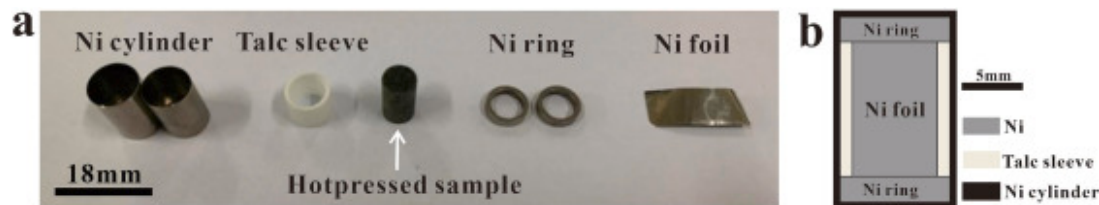
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Fig. 1. Specimen assembly of experiments for Paterson-type gas apparatus.

2.3. Experimental conditions and procedure

Using the Paterson apparatus, triaxial compression experiments were carried out on the “as is” granulite. The experiments were carried out under constant confining pressure of 300MPa, with temperatures ranging from 1273K to 1423K, and strain rates ranging from $6 \times 10^{-6} \text{ s}^{-1}$ to $1 \times 10^{-5} \text{ s}^{-1}$ (Paterson, 1970; Shao et al., 2011). In water-added experiments, the samples were subjected to a hot-pressing procedure and placed in a talc sleeve. This was done because talc undergoes dehydration reactions at high temperatures (Fig. 2), creating a hydrated environment that is conducive to sample deformation. The deformed samples were exposed to temperatures ranging from 1173K to 1373K, a constant confining pressure of 300MPa, and strain rates ranging from $6 \times 10^{-6} \text{ s}^{-1}$ to $1 \times 10^{-5} \text{ s}^{-1}$. The temperature and confining pressure were kept constant throughout each type of deformation experiment. The samples achieved a steady-state creep at each rate step, and the total strain for each strain rate was $\geq 2\%$. The totally accumulated finite strain for each sample ranged from 14% to 20%. Following the deformation experiment, the samples were recovered, cut along the axial plane, and then polished to thin pieces.



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Fig. 2. Schematic for the assembly of water-added samples. a. Materials required for assembling water-added samples; b. Schematic diagram of the assembly.

2.4. Fourier transform infrared (FTIR) analyses

In order to determine the water contents in the original and experimentally deformed mineral samples, we double-polished thin pieces taken from the samples to create a thick section that was between 100 and 120 μm thick. FTIR spectroscopic measurements were conducted using the Bruker LUMOS II instrument. An FTIR microscope equipped with a KBr [beam splitter](#) and a liquid-nitrogen-cooled MCT-A detector was used. Each spectrum was obtained by averaging over 256 scans with a resolution of 4cm^{-1} . The aperture size was adjusted to $30 \times 30 \mu\text{m}$ based on the size of the grains. Measurements were performed in regions that were optically clean, devoid of cracks, and free from inclusions. A modified version of Paterson's law (1982) was used to determine the water concentrations after more than fifteen grains of each mineral were examined.

$$c = \frac{1}{150\lambda d} \int \frac{K(v)}{3780-v} dv \quad (1)$$

$$C_{H_2O} = \frac{18c}{2 \times 10^3 \rho} \times 10^6 \quad (2)$$

Where c is the molar coefficient, $K(v)$ is the absorption coefficient, typically determined at the peak of the absorption, v is the wavenumber, d is the sample thickness (cm), λ represents the orientation factor of the OH bond in the crystal, and the samples under study are polycrystalline, hence $\lambda=1/3$. ρ is the density of the sample. The density of the samples is 2.82g/cm^3 . The baseline of each spectrum was manually adjusted, followed by the integration of the OH absorption band between 3000 and 3800cm^{-1} . The thick section's thickness was measured with a digital micrometer, and the resultant value was the average of ten measurements made throughout the section. Measurements were performed on over 15 distinct grains of each mineral to reduce

uncertainties resulting from the application of a transmitted infrared beam on unoriented anisotropic minerals ([Kovács et al., 2008](#); [Withers, 2013](#)).

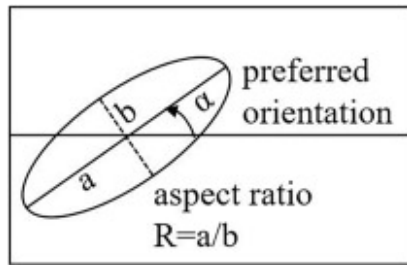
2.5. Mineral composition and microstructural analyses

The polished surfaces of the thin pieces were analyzed to gather quantitative mineral compositions using a Tescan Integrated Mineral Analyzer (TIMA) mineralogy system at Guangzhou Tuoyan Analytical Technology Co., Ltd. An acceleration voltage of 25kV and a probe current of 8.24nA were employed with a working distance maintained at 15 mm. High-resolution mapping parameters included a pixel spacing of 3μm and spot spacing of 9μm to ensure optimal spatial resolution. The current and BSE signal intensity were calibrated on a platinum Faraday cup using the automated procedure. EDS performance was checked using manganese standard. Quantitative phase analysis was performed through automated digital image processing, with mineral phase volume percentages being systematically calculated based on X-ray area scanning and standardized stoichiometric algorithms.

The microstructural observations and composition analyses were conducted using a Zeiss Sigma SEM system equipped with an Oxford X-Max N energy-dispersive X-ray spectroscopy detector with a detector size of 50mm² and an Oxford Nordlys Nano electron backscattered diffraction (EBSD) detector at the State Key Laboratory of Earthquake Dynamics, Institute of Geology, China Earthquake Administration, Beijing, China. The acceleration voltage is 20kV, and the dwell time on the sample is 2–3h. Crystal orientation images (1520–3800-μm long and 1140–2850-μm wide) were obtained with a step size of 5μm using the Oxford Instruments HKL AZtec software. The precision of the Hough transform and data acquisition conditions, such as step size, scan speed, and detector settings, can introduce angular errors (Wheeler et al., 2001; 2024).

Noise reduction was performed using CHANNEL 5 software, and missing data were filled from at least eight identical neighboring data points with similar orientations ([Bachmann et al., 2010](#)). The original calibration rates ranged from 70% to 80%. Points with mean angular deviation index of >1 were removed before grain reconstruction as well as grains <2 pixels. Grain boundaries were mapped by comparing the misorientation angle between neighboring pixels. In this context, grain boundaries were defined as misorientations ≥10° and low-angle grain boundaries as misorientations of 2–10°. The pole figures are presented using a lower-hemisphere projection with a half-width of 10. The fabric strength of the minerals is quantified by the J-index and M-index. Pole figures, J-index, and M-index are displayed and calculated based on all pixel points. Given the non-uniform grain size of the minerals in the sample, this method of calculation and presentation is more reasonable and representative, providing a rational

areal fraction weighting for the orientation distribution function (Mainprice et al., 2014). The shape-preferred orientation (SPO) was obtained using MTEX (Bachmann et al., 2010; Henry et al., 2017). The determination of the Shape Preferred Orientation (SPO) is illustrated in Fig. 3. Using the vertical compression direction as the reference line, the angle between the long axis of the grain and the reference line defines the shape preferred orientation. Simultaneously, the aspect ratio R , defined as the ratio of the long axis to the short axis, characterizes the anisotropy. A value of $R=1$ indicates equant or circular grains, while $R>1$ signifies elongated grains. Grain orientation spread was calculated using the “KAM” function that show each grain's internal misorientation.



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Fig. 3. Fabric parameters: Anisotropy R and shape preferred orientation α (the angle between the fabric's long axis and the selected reference line, which in this case is the vertical compression direction). The equivalent parameters of the strain ellipse: aspect ratio R and the angle α between the long axis of the strain ellipse and the reference line direction.

The grain sizes were determined by calculating the diameters of circles with equivalent areas, using the formula $d=2(S/\pi)^{0.5}$, where S represents the cross-sectional area of the grain. The arithmetic means were calculated as the equivalent area diameter. The grain size calculations incorporated a scaling factor of $4/\pi$, based on the assumption of spherical particle shapes (Underwood, 1970). The grain volume fraction was derived from the area fraction of the phase, as determined through image analysis. The error associated with the average grain size was assessed based on the variation in the grain size distribution, which is related to the standard deviation of the mean. This is connected to the standard deviation of the distribution via the formula $\sigma_{\text{mean}}=\sigma/N^{0.5}$, where σ denotes the standard deviation and N represents the number of observations in the sample used to estimate the mean. For clinopyroxene and plagioclase, the σ_{mean} values ranged from 0.17 to 0.28, while for olivine and orthopyroxene, these values ranged from 0.27 to 1.15. Furthermore, measuring grain sizes from EBSD data, rather than from chemically etched sections, helps

to avoid biases that may arise from the subjective identification of grain boundaries and the inherently lower resolution of optical imaging typically used with etched sections (Hansen et al., 2011).

To precisely identify the melt phase, we leverage three complementary analytical methods—EBSD, TIMA, and SEM—based on the defining characteristics of melts: 1) Absence of crystalline structure (no diffraction patterns in EBSD), 2) Complex elemental composition (detected by TIMA and SEM-EDS, including elements such as O, Na, Mg, Al, Si, K, Ca, Fe).

3. Results

3.1. Mechanical data

Four triaxial compression deformation experiments on “as is” clinopyroxene–plagioclase aggregates and five deformation experiments on water-added clinopyroxene–plagioclase aggregates were conducted respectively. To ensure accurate mechanical data, we adjusted the loading stress and displacement recorded by the instrument acquisition system using the method previously described by [Li et al. \(2023\)](#) and [Shao et al. \(2022\)](#). The deformation conditions and experimental results of all the experiments are listed in [Table 1](#). [Fig. 4a](#) and [b](#) depict the stress-strain curves. Since the sample deformation did not show significant strain hardening or softening, it was easy to determine the steady-state stress at each strain rate. Stresses drop as strain rates decrease or temperature rises under the applied experimental conditions. In the “as is” experiments, the rheological strength of the samples at 1423K is notably lower compared to other temperatures ([Fig. 4a](#)). Similarly, in the water-added experiments, the rheological strength of the samples at 1373K is also significantly lower than at other temperatures ([Fig. 4b](#)) (see [Table 2](#)).

Table 1. Experimental conditions and mechanical results of clinopyroxene–plagioclase aggregates deformed at 300MPa.

Sample	Temperature(K)	Strain rate (%) (s ⁻¹)	Stress (MPa)	Strain (%) at which stress was read	Stress exponent	C _{H2O} (ppm)	
						Pl	Cpx
HXD-11	Hot pressed	–	–	–	–	320.2	317.1
HXD-06	1273	1.32×10 ⁻⁵	369.8	4.2	4.1	616.7	663.3

Sample	Temperature(K)	Strain rate (%) (s ⁻¹)	Stress (MPa)	Strain (%) at which stress was read	Stress exponent	C _{H2O} (ppm)		
						Pl	Cpx	
HXD-11	Hot pressed	–	–	–	–	320.2	317.1	
		6.88×10 ⁻⁶	328.6	6.5				
		3.59×10 ⁻⁶	277.1	8.1				
		3.06×10 ⁻⁵	464.5	11.8				
HXD-05	1323	1.26×10 ⁻⁵	236.3	3.3	3.3	649.9	646.5	
		6.85×10 ⁻⁶	213.5	4.3				
		1.35×10 ⁻⁵	259.6	5.9				
		2.88×10 ⁻⁵	317.1	7.9				
HXD-03	1373	1.31×10 ⁻⁵	174.0	3.5	3.2	399.0	445.5	
		5.96×10 ⁻⁶	154.3	5.8				
		2.77×10 ⁻⁶	114.0	7.7				
		3.35×10 ⁻⁵	249.6	11.9				
		5.40×10 ⁻⁵	290.8	15.7				
HXD-04	1423	1.30×10 ⁻⁵	55.0	3.7	2.5	607.9	673.8	
		8.07×10 ⁻⁶	43.1	5.0				
		2.90×10 ⁻⁵	88.2	7.2				
		5.78×10 ⁻⁵	109.0	9.7				
		1.19×10 ⁻⁴	131.4	12.9				
HXW-16	1173		6.00×10 ⁻⁶	143.6	2.6	4.6	1821.3	1998.4

			1.66×10^{-5}	192.6	6.1		
			3.00×10^{-5}	204.7	10.1		
			6.15×10^{-5}	238.2	14.1		
HXW-25	1223		1.35×10^{-5}	136.4	3.0	4.7	1676.5
			3.60×10^{-5}	171.0	5.6		
			6.02×10^{-5}	205.6	10.9		
			1.21×10^{-4}	221.7	13.3		
			2.55×10^{-4}	252.0	17.6		
HXW-17	1273		1.36×10^{-5}	52.5	3.2	3.0	2199.4
			2.84×10^{-5}	100.5	7.3		
			6.06×10^{-5}	134.5	10.7		
			1.09×10^{-4}	164.7	14.7		
			2.04×10^{-4}	191.1	18.9		
HXW-13	1323		1.35×10^{-5}	32.1	3.6	2.3	1657.8
5			2.56×10^{-5}	61.4	6.9		
			5.25×10^{-5}	89.4	11.1		
			1.25×10^{-4}	118.6	14.9		
			1.96×10^{-4}	147.6	19.4		
HXW-22	1373		1.45×10^{-5}	15.3	3.4	3.9	2033.2
			2.79×10^{-5}	24.9	6.3		
			5.97×10^{-5}	33.3	9.6		

1.57×10^{-4} 37.7 13.6

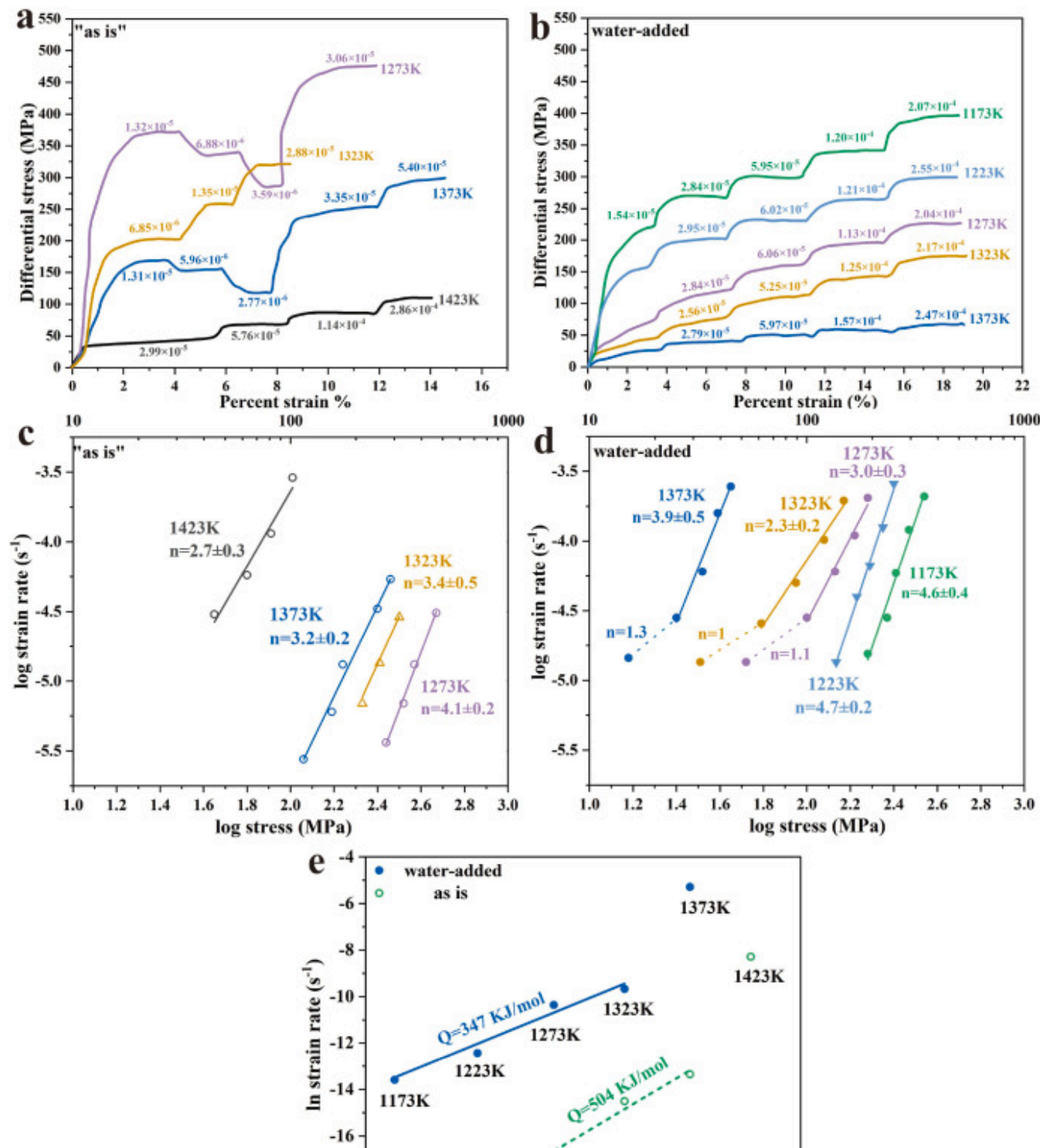
 2.47×10^{-4} 44.3 19.3

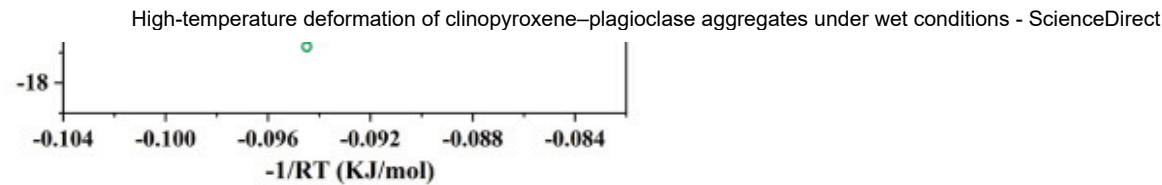
HXW-16	Not deformed	–	–	–	–	770.5	992.2
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Table 2. Mineral and melt content in hot-pressed and deformed clinopyroxene–plagioclase aggregate samples.

Sample	T/K	Pl/%	Cpx/%	Melt/%	Melt2/%	Opx/%	Ol/%
HXD-11	HP	57.16	36.20	5.38	0.23	0.21	0.07
HXD-06	1273	57.09	39.31	2.07	0.28	0.15	0.21
HXD-05	1323	58.34	38.05	1.91	0.24	0.12	0.23
HXD-03	1373	58.30	38.93	1.94	0.10	0.11	0.13
HXD-01	1423	56.52	38.82	3.83	0.08	0.06	0.02
HXW-16	1173	55.35	37.72	4.45	0.73	0.11	0.32
HXW-25	1223	55.72	35.60	7.68	0.13	0.17	0.16
HXW-17	1273	55.05	38.15	3.60	1.02	0.20	0.54
HXW-13	1323	55.60	37.63	4.11	1.00	0.32	0.50
HXW-22	1373	56.85	34.49	4.85	1.39	0.24	0.19

HXD-##, “as is” clinopyroxene–plagioclase aggregates; HXW-##, water-added clinopyroxene–plagioclase aggregates.





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Fig. 4. Mechanical results of high-temperature creep of clinopyroxene–plagioclase aggregates. A total of 9 specimens of clinopyroxene–plagioclase aggregates were subjected to triaxial compression deformation under multistage constant strain rates ranging from 6×10^{-6} to $1 \times 10^{-5} \text{ s}^{-1}$. The experiments were conducted at temperatures between 1173 and 1423K, with a confining pressure of 300MPa. The samples achieved steady-state creep at each strain rate, with a minimum total strain of 2%. Stress–strain curves for (a) “as is” and (b) water-added clinopyroxene–plagioclase aggregates. (c) The plot of log-log strain rate vs. stress for the steady-state flow of “as is” clinopyroxene–plagioclase aggregates at 1,273, 1,323, 1,373, and 1423K, respectively, yielding stress exponents of $n=4.1, 3.4, 3.2,$ and 2.7 ; (b) plot of log-log strain rate vs. stress for the steady-state flow of water-added clinopyroxene–plagioclase aggregates at 1,173, 1,223, 1,273, 1,323, and 1373K, respectively, yielding stress exponents of $n=4.6, 4.7, 3.0, 2.3,$ and 3.9 ; the stress exponent (n) is 1 in the lower stress; (c) plot of $\ln$ strain rate vs. inverse temperature for normalized stress of 100MPa, yielding activation energy of $Q=504 \text{ kJ/mol}$ for “as is” clinopyroxene–plagioclase aggregates, $Q=347 \text{ kJ/mol}$ for water-added clinopyroxene–plagioclase aggregates.

The strain rate ($\dot{\epsilon}$) can be expressed by a function of differential stress (σ) and temperature (...) using a steady-state flow law of equation (3):

$$\dot{\epsilon} = A \sigma^n \exp \left(-\frac{Q}{RT} \right) \quad (3)$$

Where, A is a material-dependent parameter, n is the stress exponent, Q is the activation energy, R is the gas constant, and T is the absolute temperature. The stress exponent can be computed by determining the slope of the stress corresponding to the strain rate on a

log-log plot (Fig. 4c and d). For the experiments carried out under “as is” conditions, within the temperature range of 1273K to 1423K, the stress exponent (n) for clinopyroxene–plagioclase aggregates ranges from 2.7 to 4.1, with an average of 3.6 ± 1.6 (Fig. 4c). In the experiments with added water, conducted at temperatures of 1173–1373K, the stress exponent (n) of 2.3–4.7, with an average of 3.7 ± 1.0 (Fig. 4d). It is noteworthy that at temperatures of 1,273, 1,323, and 1373K, under conditions of relatively low stress, the stress exponent (n) takes on the value of 1.

The determination of the activation energy involved in plotting the logarithm of the inverse temperature against the strain, with the stress normalized to 100MPa based on the n value (Fig. 4e). In the “as is” clinopyroxene–plagioclase aggregates experiments, a significant decrease in rheological strength at 1423K was observed. Utilizing data obtained at 1273–1373K, the Q value was calculated as 504 ± 85 kJ/mol. In the water-added experiments, a noticeable decrease in rheological strength at 1373K was observed, the Q value was then calculated to be 347 ± 48 kJ/mol using data from the temperature range of 1173K to 1323K. It is worth noting that the activation energy observed in the water-added experiments is significantly lower compared to that observed in the “as is” experiments.

Thus, the rheological behavior of clinopyroxene–plagioclase aggregates under two different water content conditions can be described by the following constitutive equation:

$$\text{“As is” samples : } \dot{\epsilon} = 10^{6.7 \pm 0.7} \sigma^{3.6 \pm 0.6} \exp \left(-\frac{504 \pm 85 \text{ KJ/mol}}{RT} \right) \quad (4)$$

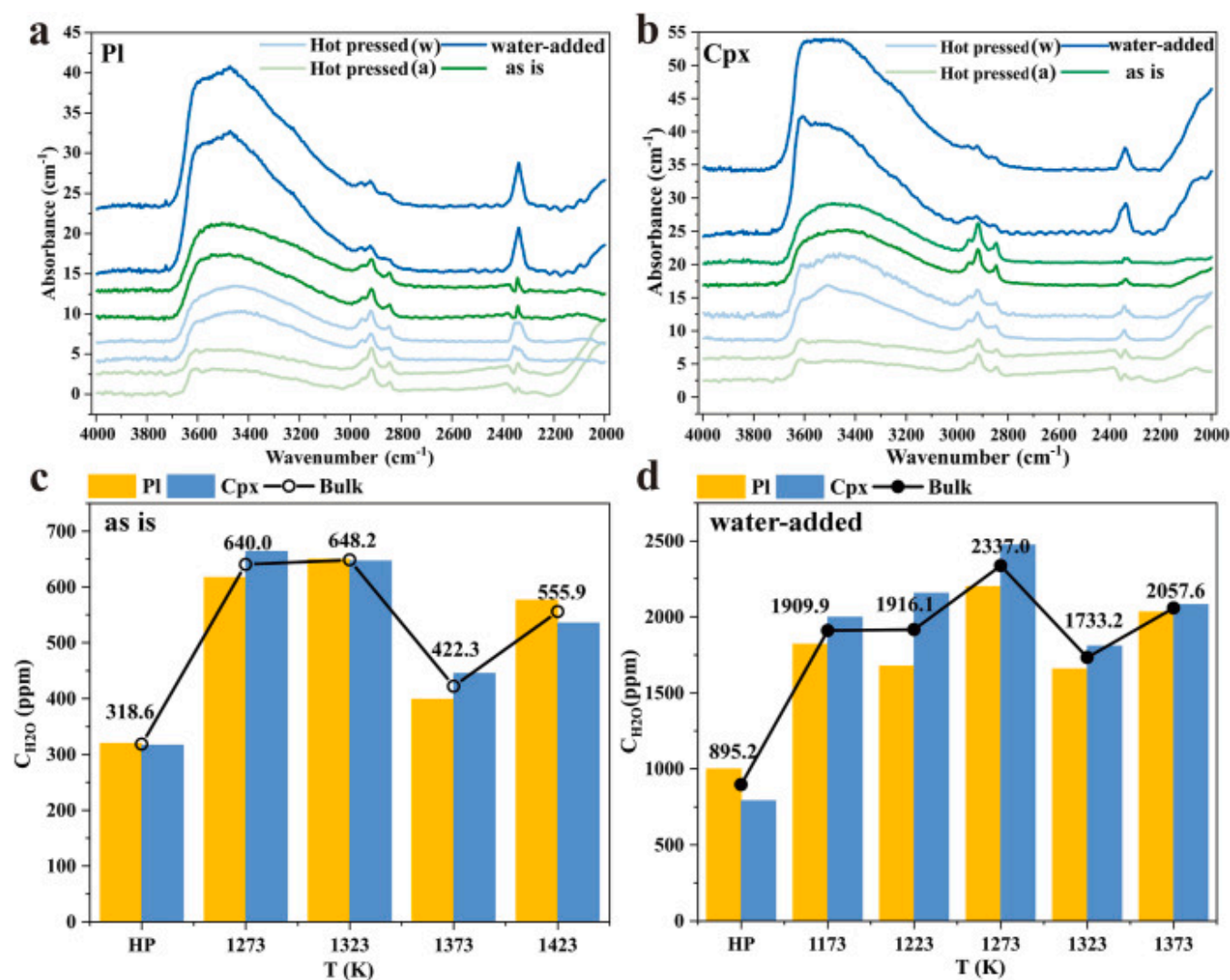
$$\text{Water-added samples: } \dot{\epsilon} = 10^{2.3 \pm 0.4} \sigma^{3.7 \pm 1.0} \exp \left(-\frac{347 \pm 48 \text{ KJ/mol}}{RT} \right) \quad (5)$$

where $\dot{\epsilon}$ is in s^{-1} , σ in MPa, and in Kelvin.

3.2. Water concentration

Fig. 5 displays the typical infrared spectra of plagioclase and clinopyroxene. The spectra include samples that were “as is” as well as water-added samples, before and after deformation. All spectra were standardized to a thickness of 1 cm. The FTIR data are summarized in Table 1. Most FTIR spectra of plagioclase show, a distinct and intense band is observed at $3570\text{--}3610\text{cm}^{-1}$, along with a broader band centered around 3500cm^{-1} (Fig. 5a). According to Johnson and Rossman (2004), these bands are assumed to be associated with the presence of structural OH groups or water molecules. In the case of clinopyroxene, the FTIR spectra exhibit

a sharp peak of around 3620cm^{-1} and a broader peak of around 3530cm^{-1} , which is likely attributed to the stretching vibration of OH (Fig. 5b) (Bell and Rossman, 1992; Ingrin and Skogby, 2000).



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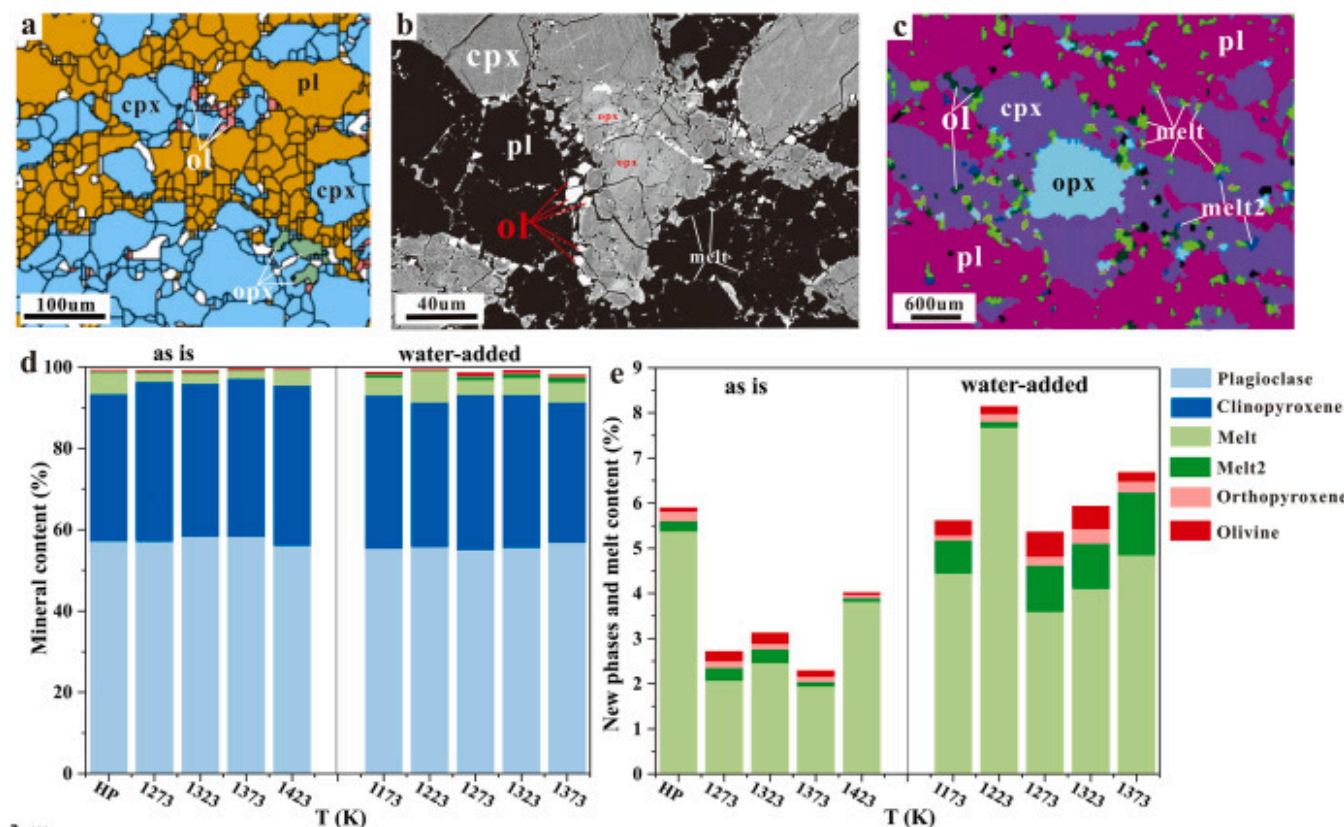
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Fig. 5. Representative FTIR spectra (a, b) and water contents (c, d) in clinopyroxene (Cpx) and plagioclase (Pl) of the hot-pressed and deformed samples. All spectra were normalized to 1 cm thickness. (a) and (w) mean “as is” and “water added”.

In the “as is” clinopyroxene–plagioclase aggregates, the water content in clinopyroxene and plagioclase is approximately the same. We calculated the overall water content of the samples based on average data from two major minerals. As depicted in [Fig. 5c](#), the water content of the hot-pressed sample is 318.6ppm. However, in the deformed samples, the water content is around 600ppm, which is consistent with the data reported for natural clinopyroxene–plagioclase aggregates in eastern China (150–950ppm). In the water-added clinopyroxene–plagioclase aggregates ([Fig. 5d](#)), the water content in the hot-pressed samples (895.2ppm) is significantly lower than that in the deformed samples (~2000ppm), indicating dehydration of the talc that provided a hydrated environment during the deformation process.

3.3. Partial melt and mineral reaction

The mineral composition of the clinopyroxene–plagioclase aggregates was analyzed using EBSD, SEM, and TIMA techniques ([Fig. 6a, b, and 6c](#)). The original minerals in the initial samples consisted of plagioclase and clinopyroxene. However, new phase minerals of fine-grained orthopyroxene and olivine, as well as melt, were observed along the grain boundaries in the hot-pressed and deformed samples ([Fig. 6, Fig. 14](#)). The hot-pressed samples had a melt content of ~5.6%, while the deformed “as is” samples had a melt content ranging from 2.2% to 3.9% ([Fig. 6d–e, Table 2](#)). This indicates that some of the melt crystallized into olivine and clinopyroxene during deformation. Alternatively, the water-added samples exhibited a higher melt content of ~4.6%–9.1% in the deformed samples, compared to the “as is” samples ([Fig. 6d–e, Table 2](#)), which indicates that the higher melt content because of the higher water content.



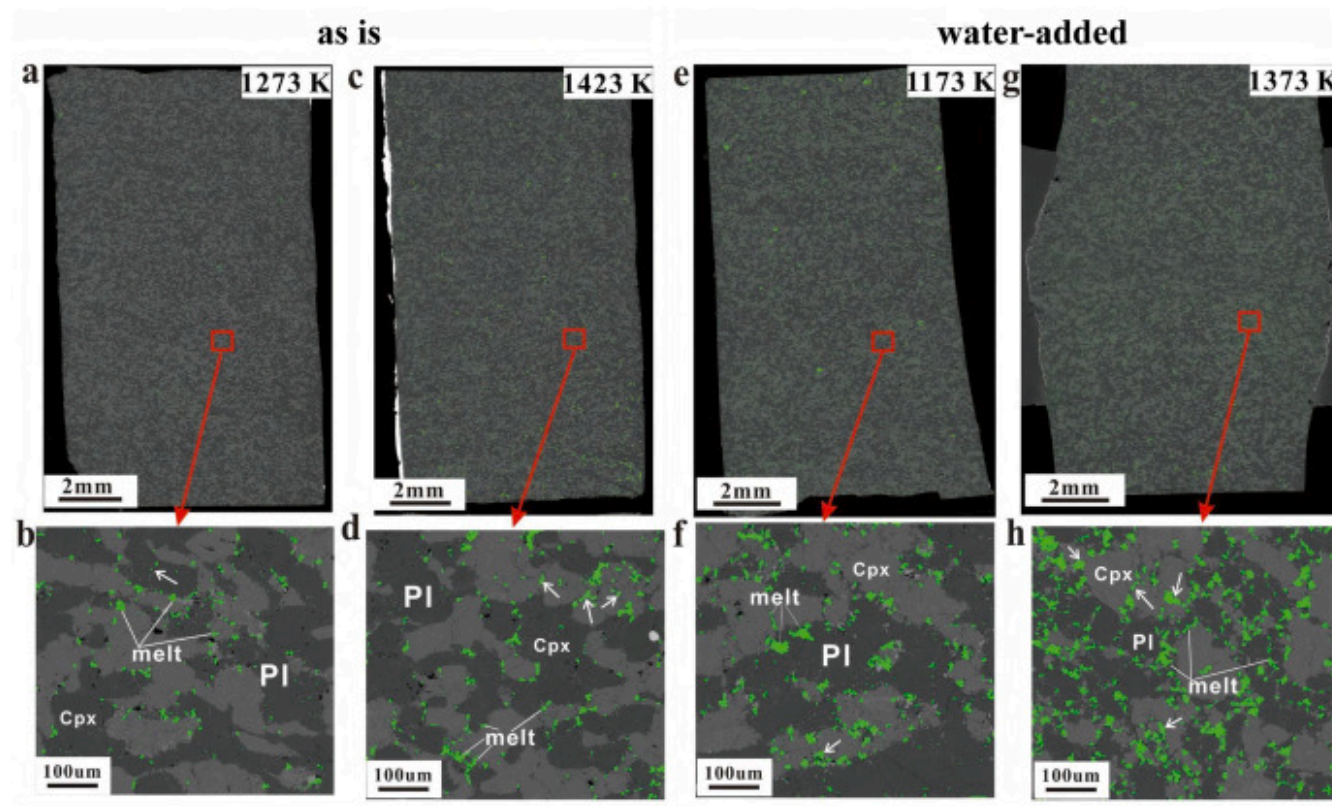
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Fig. 6. The character of melt and reaction newly formed minerals in the clinopyroxene–plagioclase aggregates samples. The original phases comprise plagioclase and clinopyroxene, while the new phases comprise orthopyroxene and olivine. (a) the phase map generated with EBSD, (b) the BSE image made with SEM, and (c) the phase map made with TIMA. (d) Volume content of mineral phases in experimental clinopyroxene–plagioclase aggregates samples. (e) Volume content of melt and newly formed phases in deformed samples. The volume fraction of melt and new phases in the deformed water-added samples is higher than that in the deformed “as is” samples, with no clear trend observed with increasing temperature.

We identified two types of melts, one containing Mg (Melt) and one without Mg (Melt 2) (Fig. 6c). Most of the melt with Mg is distributed at the triple junctions of the grains (Fig. 7), while a smaller amount of melt is found along the grain boundaries (Fig. 7,

Fig. 14). Furthermore, as the temperature rises, the melt content increases (Fig. 7, Fig. 14). Additionally, we observed that the connectivity of the melt forms a network-like structure at 1423K in the “as is” granulite (Fig. 7c and d), whereas this connectivity occurs at a lower temperature of 1373K in the water-added granulite (Fig. 7g and h). This suggests that the presence of water promotes the occurrence of melt connectivity at lower temperatures.



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Fig. 7. The distribution of melt in “as is” clinopyroxene–plagioclase aggregates at 1,273K (a, b) and 1,423K (c, d) and water-added clinopyroxene–plagioclase aggregates at 1,173K (e, f) and 1,373K (g, h). The melt film appeared at the two phases and single phase (white arrows) grain boundaries of minerals.

The content of newly formed olivine and orthopyroxene in the hot-pressed samples is 0.28%, and that in the “as is” deformed samples while it ranges from 0.29% to 0.36% in at temperatures of 1273–1373 K (Fig. 6e), which is similar to that of the hot-pressed samples. However, the content of newly formed minerals is approximately 0.33%–0.82% in the deformed water-added samples, which is higher compared to the hot-pressed and “as is” deformed samples. These findings indicate that the proportion of newly formed minerals increases with an increase in water content, and deformation promotes the formation of new-phase minerals.

3.4. Grain size

In the hot-pressed samples, the grain size distribution of the plagioclase and clinopyroxene, remains relatively uniform, with an average grain size of $\sim 17\mu\text{m}$ and 2–5 vol% porosity. A significant proportion of grains (0–20 μm) constitutes $\sim 75\%$ of the total grain count but contributes only 20% of the total volume. Though fewer in number, larger grains ($>20\mu\text{m}$) occupy a substantially greater volume. Due to the relatively low statistical count of the newly formed minerals, olivine and orthopyroxene, their grain size distribution is less reliable. However, their grain sizes appear to be predominantly small, mostly below 20 μm (Fig. 8).

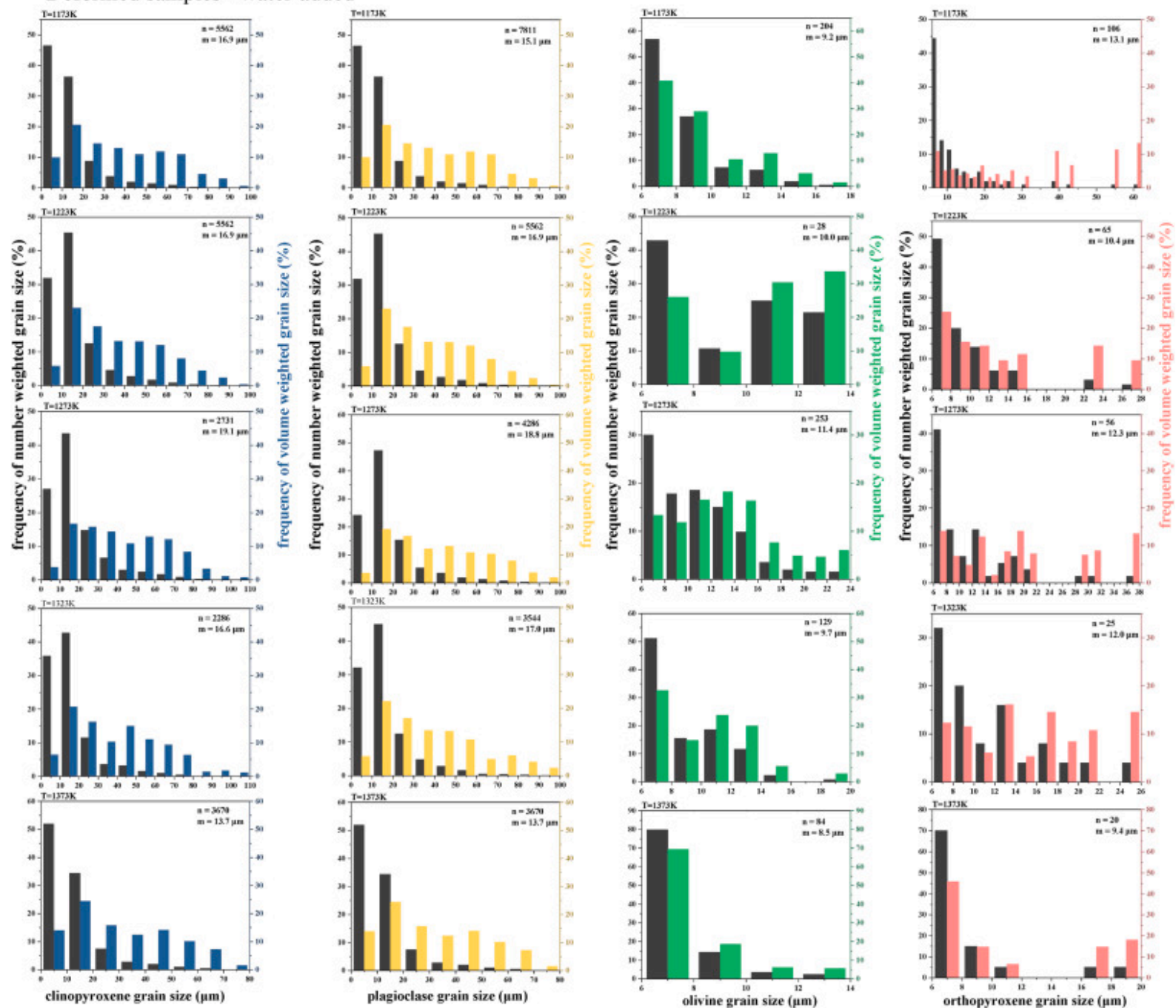


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Fig. 8. Number-weighted and volume-weighted histograms of grain sizes for the undeformed sample and for the deformed “as is” samples at temperature of 1273–1423K, with arithmetic means (m) for clinopyroxene, plagioclase, olivine and orthopyroxene indicated. Black histograms: Number-weighted grain-size distributions for clinopyroxene, plagioclase, olivine and orthopyroxene. Blue, yellow, green and red histograms: Volume-weighted grain size distributions for clinopyroxene, plagioclase, olivine and orthopyroxene, respectively. n is the number of grains analyzed. m is the arithmetic means. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

In “as is” deformed samples, plagioclase and clinopyroxene exhibit congruent grain size distributions. Particles within the 1–20 μm range dominate numerically ($\sim 75\%$) but represent only $\sim 25\%$ of the total volume fraction. Neoformed olivine and orthopyroxene phases are predominantly characterized by $< 12\mu\text{m}$ grains. Notably, at 1423K, both the abundance and volumetric contribution of fine-grained particles ($< 20\mu\text{m}$) increased significantly compared to those formed at 1273–1373K ([Fig. 8](#)). In water-added deformed samples, plagioclase and clinopyroxene display analogous grain size distributions, with 1–20 μm grains remaining numerically dominant yet volumetrically minor. At 1373K, these systems exhibited a pronounced enhancement in both grain population density and volumetric proportion relative to samples deformed at 1173–1323K ([Fig. 9](#)).

Deformed samples - water-added



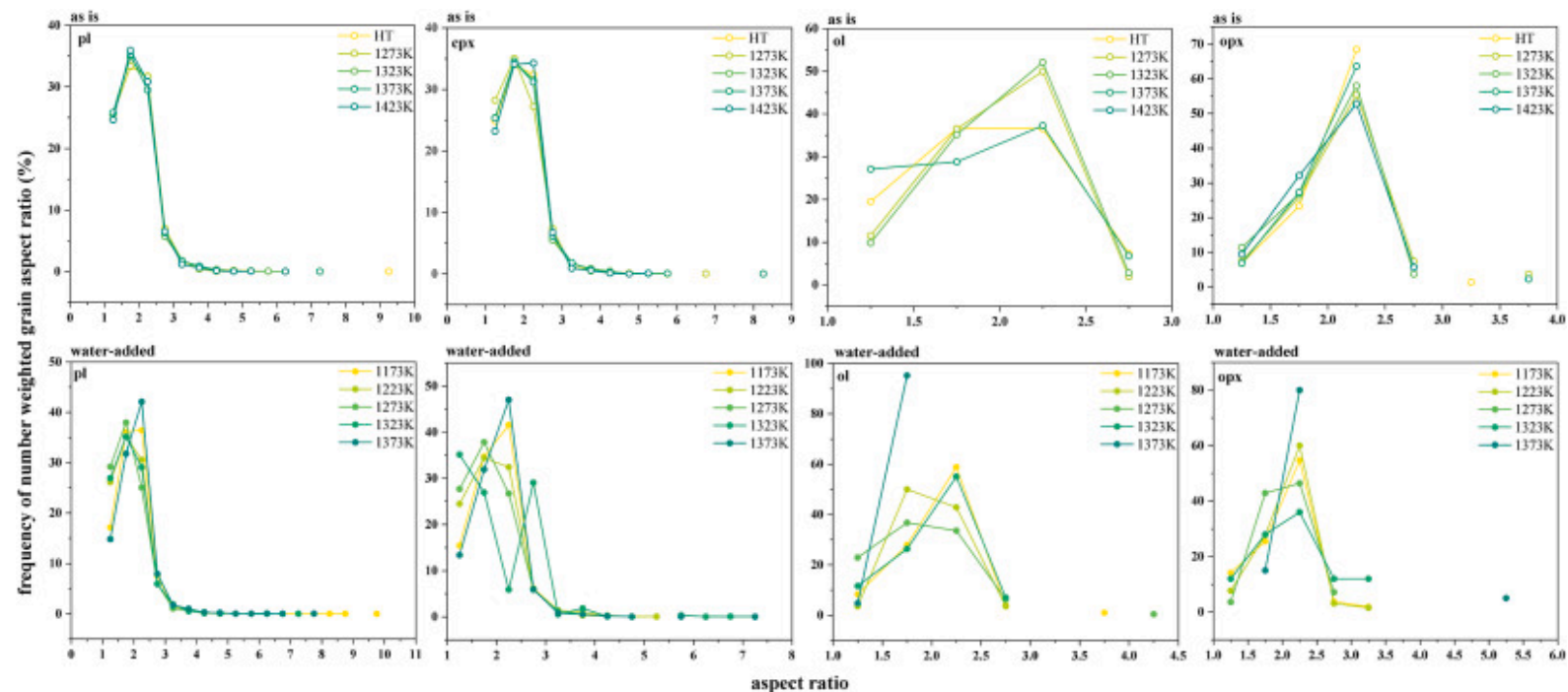
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Fig. 9. Number-weighted and volume-weighted histograms of grain sizes for the deformed water-added samples at temperature of 1173–1373K, with arithmetic means (m) for clinopyroxene, plagioclase, olivine and orthopyroxene indicated. Black histograms: Number-weighted grain-size distributions for clinopyroxene, plagioclase, olivine and orthopyroxene. Blue, yellow, green and red histograms: Volume-weighted grain size distributions for clinopyroxene, plagioclase, olivine and orthopyroxene, respectively. n is the number of grains analyzed. m is the arithmetic means. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.5. Microstructures

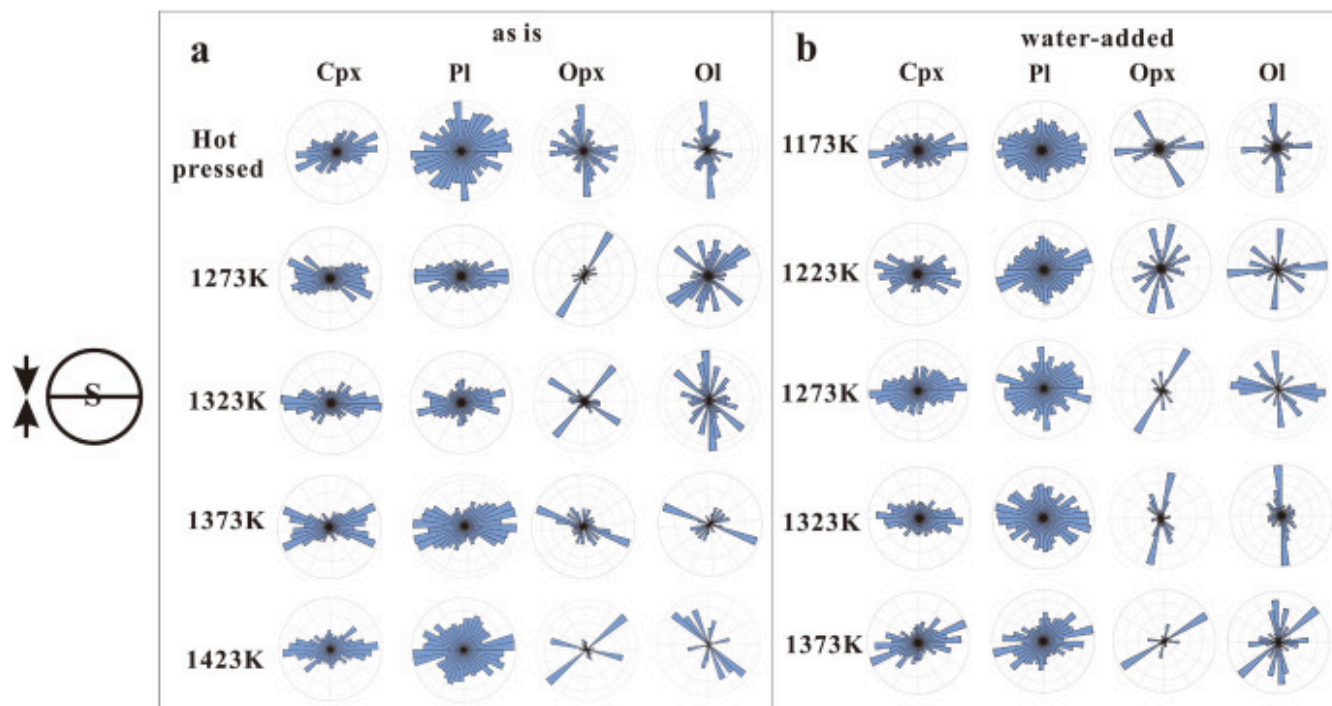
In the hot-pressed samples, clinopyroxene and plagioclase show a random distribution, indicating high sample homogeneity. The samples contain ~2vol% porosity, and both clinopyroxene and plagioclase display irregular embayed grain boundaries. Additionally, we observed some fine-grained particles and melt distributed along the grain boundaries (Fig. 14). Clinopyroxene and plagioclase grains predominantly exhibit aspect ratios ($R = \text{long/short axis}$) > 1.5 , accounting for ~75% of the population, indicating a strong prevalence of elongated grain morphologies. Similarly, neoformed olivine and orthopyroxene grains display even higher elongation, with ~80% exceeding an aspect ratio of 1.5 (Fig. 10). Clinopyroxene exhibited a SPO perpendicular to the compression direction, while plagioclase showed a random distribution (Fig. 11). Furthermore, the clinopyroxene exhibited a weak CPO, with the [100] oriented along the compression direction and the [001] axis aligned along the lineation direction. In contrast, the [100] axis of the plagioclase grains, although less distinct, appeared to be nearly perpendicular to the compression direction (Fig. 12).



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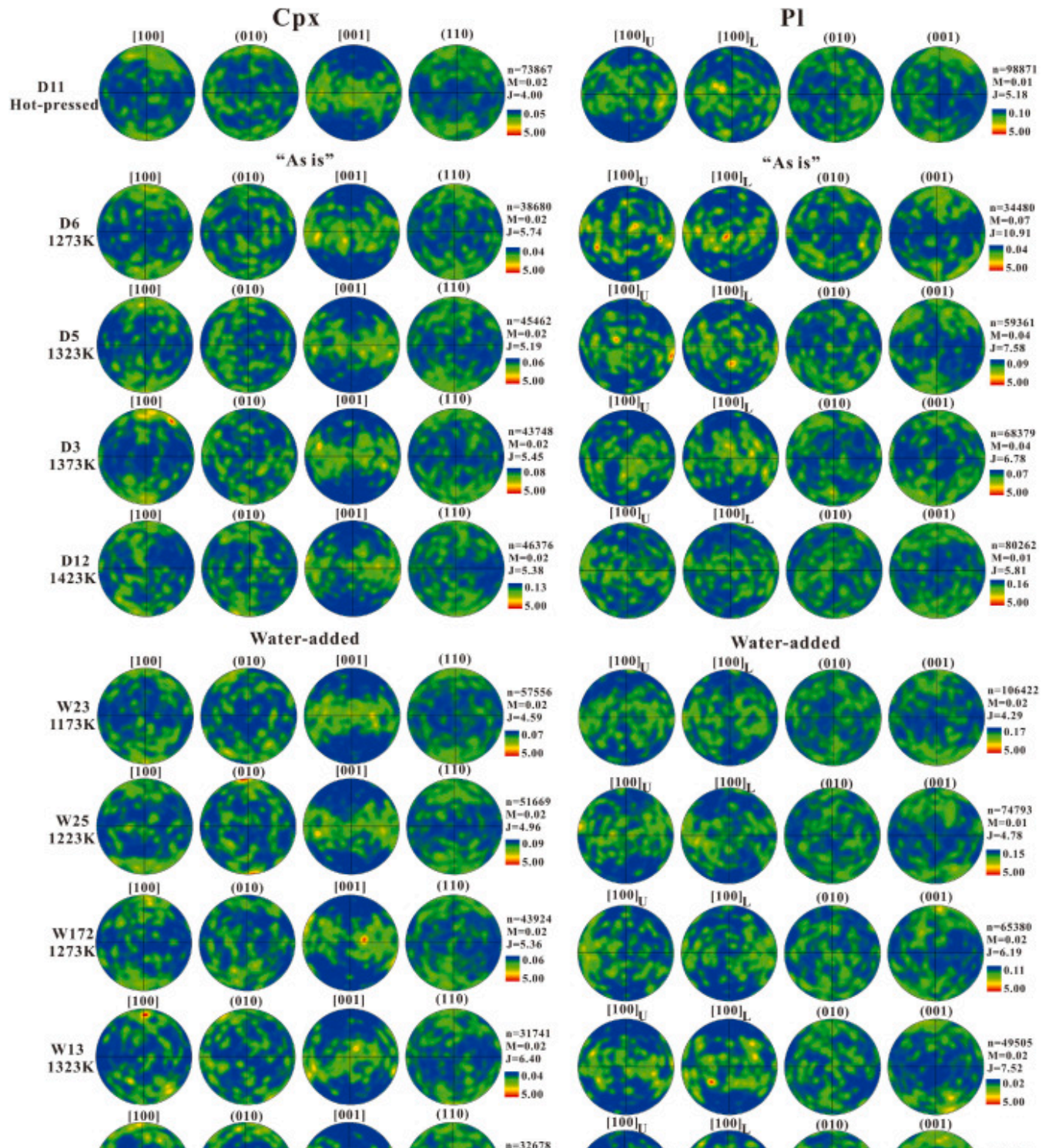
Fig. 10. Number-weighted and volume-weighted line graph of aspect ratio in clinopyroxene, plagioclase, olivine and orthopyroxene for the undeformed sample, for the deformed “as is” samples at temperature of 1273–1423K and for the deformed water-added samples at temperature of 1173–1373K.

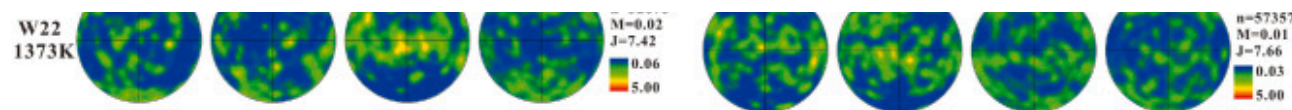


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Fig. 11. Shape-preferred orientation (SPO) maps of (a) hot pressed, “as is” samples deformed at 1273–1423K, (b) water-added samples deformed at 1173–1423K.



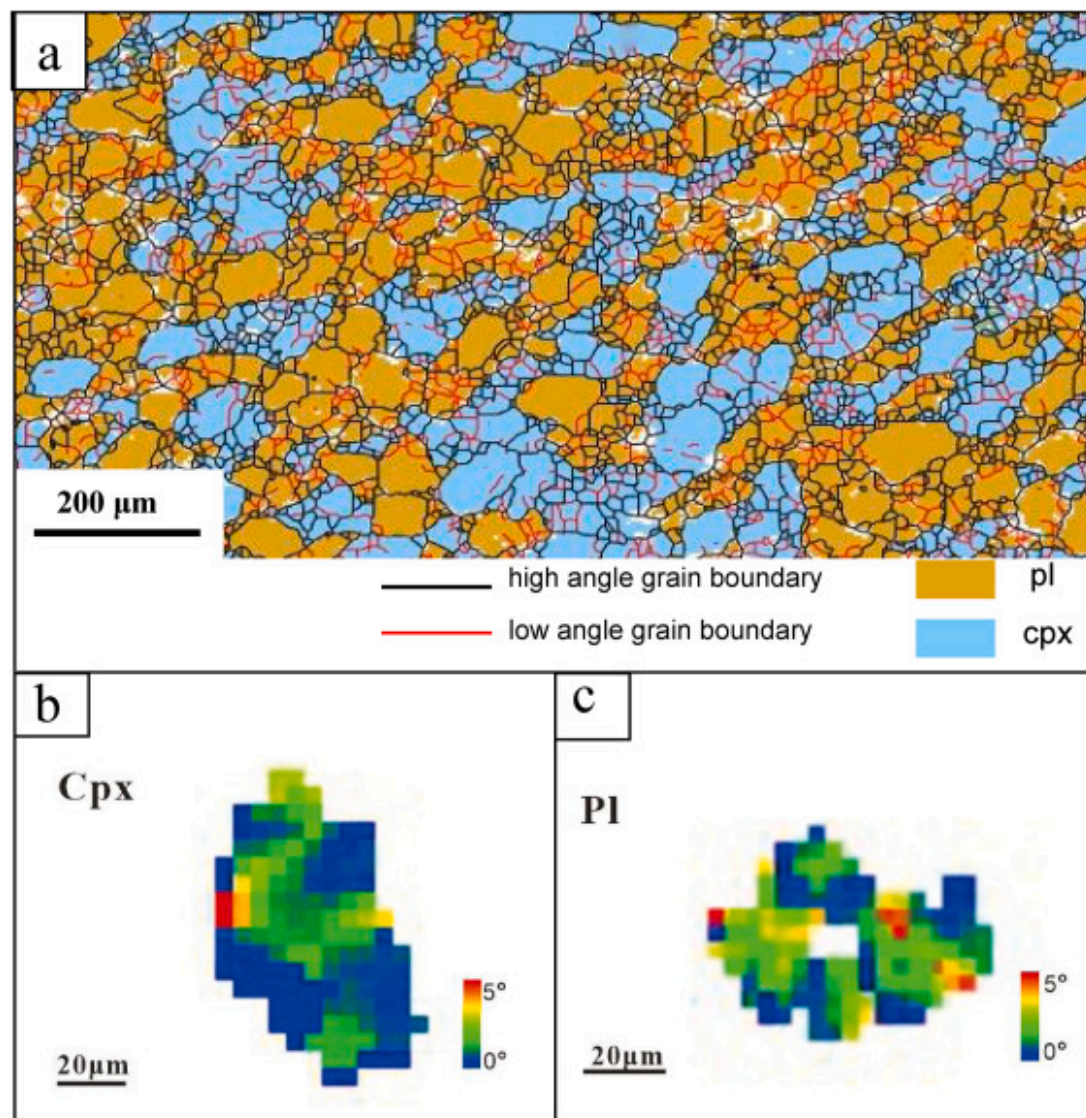


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Fig. 12. Crystallographic Preferred Orientation (CPO) maps of hot pressed, “as is” and water-added clinopyroxene–plagioclase aggregates deformed at 1173–1423K, pole figures of equal areas in the lower hemisphere. The strength of CPO is quantified by the misorientation index (M-index) developed by Skemer et al. (2005) and the J-index. The M-index is 0 for a random fabric and 1 for a single crystal, while the J-index is 1 for a random distribution and infinity for a single crystal orientation.

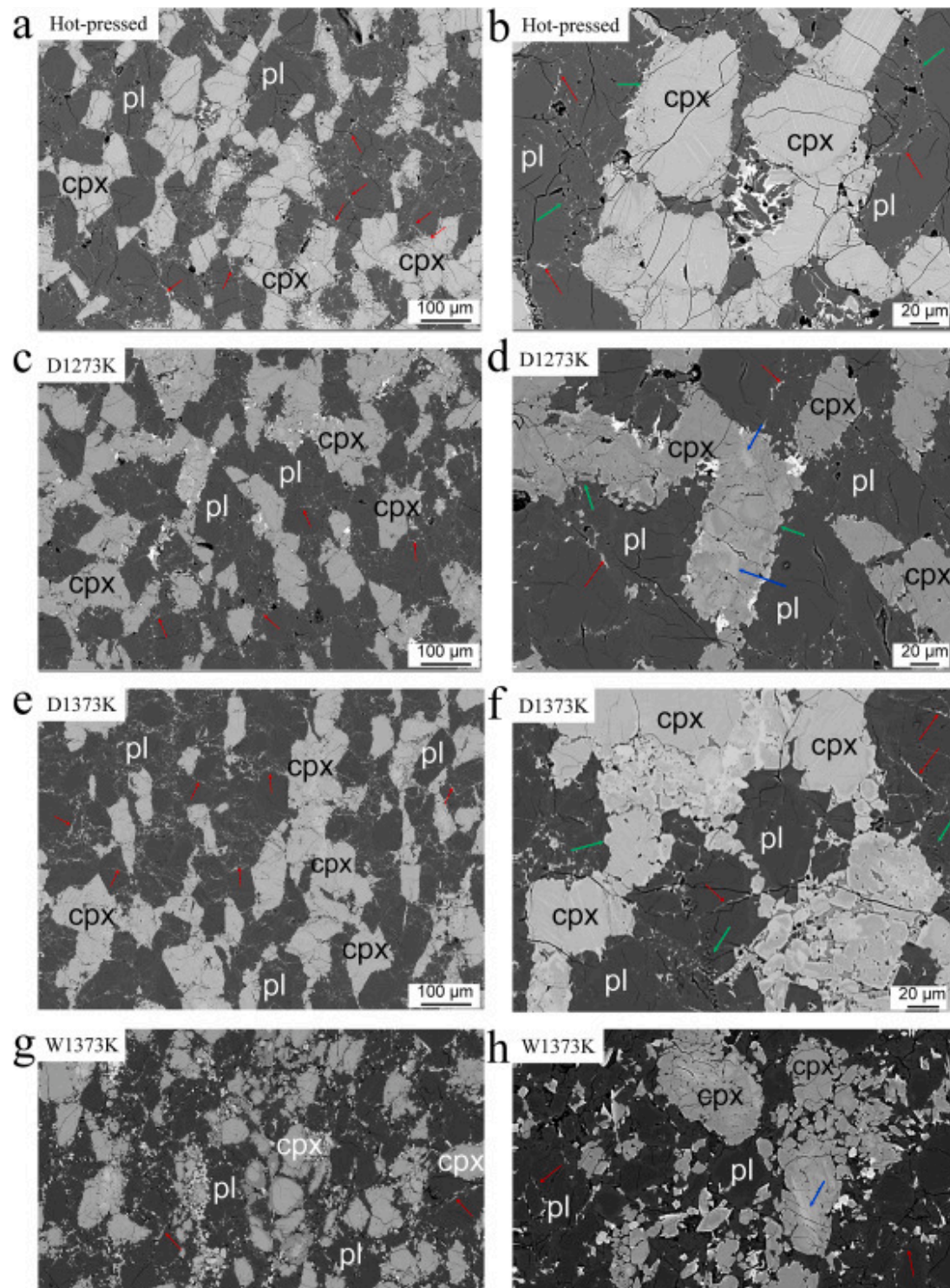
In the “as is” and water-added deformed samples, the aspect ratio distribution of the primary minerals (plagioclase and clinopyroxene) and the newly formed minerals (olivine and orthopyroxene) is largely consistent with that of the hot-pressed minerals, with most grains exhibiting an aspect ratio greater than 1.5. The distribution of aspect ratios for clinopyroxene and plagioclase remained consistent across different levels of water content and temperature (Fig. 10). In contrast to the hot-pressed samples, the clinopyroxene and plagioclase in the “as is” and water-added deformed samples exhibited more pronounced SPO (Fig. 11). The SPO of clinopyroxene and plagioclase weakened with higher water content (Fig. 11b). However, the SPO of olivine and orthopyroxene appeared to be random. The EBSD analysis revealed that the CPO of clinopyroxene and plagioclase in the deformed samples exhibited a more pronounced CPO compared to the hot-pressed samples (Fig. 12). Furthermore, the development of CPO was found to be stronger in samples with higher water content (Fig. 12). In the “as is” samples, the CPO development showed an increase with rising temperature of 1273–1373K, but decreased at 1,423K. Similarly, in the water-added samples, the CPO development increased with rising temperature of 1173–1323K but decreased at 1,373K (Fig. 12). EBSD mapping shows that clinopyroxene and plagioclase commonly show distinct subgrain boundaries (Fig. 13a) and subgrain structures (Fig. 13b).



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Fig. 13. EBSD characterization of microstructures. (a) EBSD phase map with grain boundaries. EBSD mappings showing intracrystalline deformation features (subgrain structure) in (b) clinopyroxene and (c) plagioclase. EBSD, electron backscatter diffraction.





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Fig. 14. SEM backscatter images (a–b) of the hot-pressed sample, (c–d) of the “as is” sample deformed at 1273K, (e–f) of the “as is” sample deformed at 1373K and (g–h) of the water-added sample deformed at 1373K. Red arrows indicate melt films. Green arrows indicate the bay-like structures. The blue arrows indicate chemical reactions occurring. pl, plagioclase; cpx, clinopyroxene. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4. Discussion

4.1. Deformation mechanisms and rheological parameters

(1) Deformation mechanisms under different water contents

Microstructural observations reveal that clinopyroxene and plagioclase in deformed “as is” and water added samples exhibit SPO (Fig. 11) and CPO (Fig. 12) as well as subgrain boundaries by misorientation, coupled with significant intragranular plasticity (Fig. 13). These features collectively indicate deformation via dislocation creep. The limited finite strain (<20%) and weaker SPO and CPO in hot-pressed sample, which suggest that passive rotation of grains also contributed to SPO/CPO development in the deformed samples. However, the predominance of intragranular plasticity in coarse-grained domains (constituting ~80vol%) underscores dislocation creep as the primary deformation mechanism. The presence of melt, neofomed phases (olivine+orthopyroxene) and abundant fine-grained particles (<10μm) all show subsidiary dissolution-precipitation creep. In addition, the mechanical data obtained from the samples of deformed clinopyroxene–plagioclase aggregates revealed a stress exponent (n) of 3.6–3.7, indicating typical dislocation creep behavior.

The EBSD data showed that the CPO of clinopyroxene exhibits a [100] axis parallel to the compression direction and a [001] axis perpendicular to the compression direction. This suggests the presence of a primary slip system of (100)[001] in this study (Fig. 12). This CPO is similar to that observed in clinopyroxene subjected to dislocation creep under natural and experimental conditions (Kanagawa et al., 2008; Mauler et al., 2000; Zhang et al., 2006).

Furthermore, the CPO of plagioclase shows a [100] alignment with the lineation and a (001) alignment with the foliation, indicating the activation of the (001)[100] slip system (Fig. 12). This slip system is widely recognized to be activated during dislocation creep in plagioclase, as evidenced by natural and experimentally deformed rocks (Heidelbach et al., 2000; Ji et al., 2004; Mehl and Hirth, 2008; Wang et al., 2012; Xie et al., 2003). Ji et al. (2000) proposed that the slip direction along the [001] plane is more prevalent at low temperatures, while the slip direction along the [100] plane is more common at high temperatures. Moreover, the presence of water content can cause a transition from the [001] slip direction to the [100] slip direction. The investigation conducted in this study focused on the slip system of plagioclase within a temperature range of 1000°C to 1150°C and with water content reaching up to approximately 600ppm (Fig. 5c). The results indicated that higher temperatures and increased water content enhance the activation of the [100] slip in plagioclase. Comparatively, CPO strength, as measured by the M value and J value, of plagioclase in the deformed samples generally exceeded that of clinopyroxene under similar water content. This suggests that plagioclase is a weaker phase that is more susceptible to deformation than clinopyroxene under the same conditions.

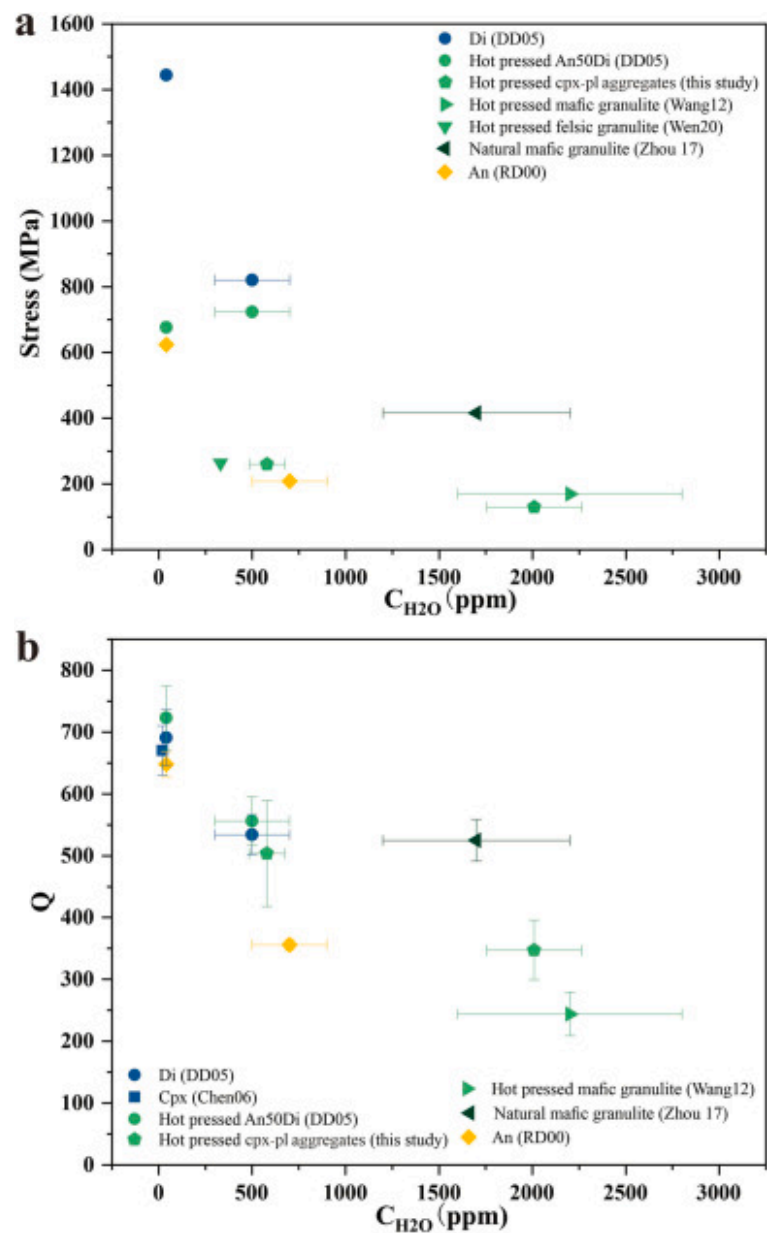
Concurrently, the presence of melt and neoformed phases (olivine+orthopyroxene) attests to localized mass transfer processes, while abundant fine-grained particles (<10µm) under higher temperature, and the presence of more melt and newly formed minerals in the samples with higher water content (Fig. 6e). The reaction to form fine-grained new phases were found in plagioclase-olivine (de Ronde et al., 2005), clinopyroxene–plagioclase (Mansard et al., 2020; Marti et al., 2017,2018, 2020; Nevskaya et al., 2025), and wet plagioclase aggregates (Baïssset et al., 2024), which induce dissolution-precipitation creep are main process. All imply subsidiary dissolution-precipitation creep played an important role under higher temperature conditions in this study, and water promotes dissolution-precipitation creep.

(2) Rheological parameters under different water contents

To analyze the impact of water on activation energy (Q) and stress, we conducted a comparative analysis with previously published data involving clinopyroxene (Dimanov and Dresen, 2005), plagioclase (Rybacki and Dresen, 2000), hot-pressed

granulite ([Wang et al., 2012](#); [Wen et al., 2021](#)), and natural granulite ([Zhou et al., 2017](#)).

Stress was normalized at a strain rate of 10^{-14} and a temperature of 1323 K ([Fig. 15a](#)). Anorthite, being the predominant mineral in the lower crust, exhibits lower strength than pyroxene in dry and wet conditions. The stress in hot-pressed clinopyroxene–plagioclase aggregates was consistently lower than in clinopyroxene but higher than in plagioclase, with similar water content conditions. The weakening effect of water is more pronounced in clinopyroxene compared to anorthite ([Dimanov and Dresen, 2005](#)). Stress in hot-pressed granulite showed a significant decrease when water content reached ~600 ppm but remained relatively constant between water contents of 600–2000 ppm. This indicates that once the water content reaches a certain level, its weakening effect diminishes. Natural rocks ([Zhou et al., 2017](#)) exhibited significantly higher stress compared to hot-pressed granulite under similar water content conditions, possibly due to their complex composition, uneven grain size, and factors such as brittle failure in natural granulite.



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Fig. 15. The (a) stress (normalized to a temperature of 1,323K and a strain rate of 10^{-14}) and (b) activation energy (Q) with varying water content. DD05 ([Dimanov and Dresen, 2005](#)), RD00 ([Rybacki and Dresen, 2000](#)), Wang12 ([Wang et al., 2012](#)), Wen20 ([Wen et al., 2020](#)), Zhou17 ([Zhou et al., 2017](#)).

The activation energy of dislocation creep under different water content is shown in [Fig. 15b](#). The activation energies of anorthite and clinopyroxene exhibit a significant decrease as the water content increases. However, the impact of water on anorthite is more pronounced than on diopside, which can be attributed to the varying water contained in these two minerals. FTIR measurements have indicated that plagioclase contains a higher trace of water compared to clinopyroxenes ([Beran, 1987](#); [Bell and Rossman, 1992](#); [Hofmeister and Rossman, 1985](#); [Ingrin et al., 1991](#); [Wilkins and Sabine, 1973](#)). In the case of hot-pressed granulite and two-phase (diopside and anorthite) aggregates, the activation energy decreases with an increase in water content. Furthermore, under the same water content, the activation energy of hot-pressed granulite and two-phase (clinopyroxene and plagioclase) aggregates closely resembles that of clinopyroxene.

(3) Effect of temperature on rheological properties and deformation mechanisms

In the “as is” clinopyroxene–plagioclase aggregate samples, we observed that the stress at 1423K was noticeably lower compared to other temperatures ([Fig. 4a](#) and [c](#)). Similarly, in the water-added experiments, the samples exhibited significantly lower stress at 1373K compared to other temperatures ([Fig. 4b](#) and [d](#)). This observation suggests that there is a correlation between temperature and the deformation mechanism, as the stress decreases with increasing temperature.

In the deformed “as is” and water-added samples, the development of CPO increases with increasing temperature at 1273–1373K ([Fig. 12](#)), suggesting that the rate of dislocation glide intensifies with temperature. However, the development of the (100)[001] slip system in clinopyroxene at 1423K is weaker compared to 1273–1373K. Moreover, the sample at 1423K exhibits weaker SPO ([Fig. 11](#)), higher melt content ([Fig. 6e](#)), more small particles ($<10\mu\text{m}$) ([Fig. 8](#)), and interconnected melts ([Fig. 7d](#)), as well as a lower stress exponent value ($n=2.7$), indicating that dissolution-precipitation creep is becoming progressively more significant.

(4) Effect of melt on rheological properties and deformation mechanisms

According to [Acosta-Vigil et al. \(2006\)](#), [Koekpe et al. \(2004\)](#), [Kvassnes and Grove \(2008\)](#), [Schulmann et al. \(2008\)](#), melting and melt transportation play critical roles in the deformation processes that occur in the lower crust, upper mantle, and during

contact metamorphism. It is believed that partial melting significantly impacts the mechanical behavior of rocks (Dell'Angelo et al., 1988; Koepke et al., 2004; Rushmer, 1995; Rutter and Neumann, 1995; van der Molen and Paterson, 1979; Vanderhaeghe, 2009; Yoshinobu and Hirth, 2002; Yoshino et al., 2009; Zhu et al., 2011). In the lower crust or upper mantle, during creep, the weakening effect caused by partial melting can result in strain localization. Additionally, under high-strain conditions, the expulsion or removal of melt from triple or quadruple junctions and grain boundaries can give rise to discrete banded structures and melt channels, thereby further intensifying the weakening effect of the melt (Kohlstedt and Holtzman, 2009; Kohlstedt et al., 2010; Rybacki et al., 2008, 2010). However, at a critical melt percentage (CMP) of ~ 0.07 , melt connectivity increases markedly, resulting in a sharp decrease in rock strength. When the melt content is below this threshold, the weakening effect of melt is negligible (Rosenberg and Handy, 2005; Holtzman et al., 2016).

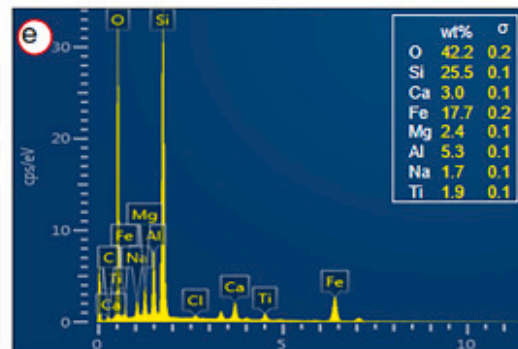
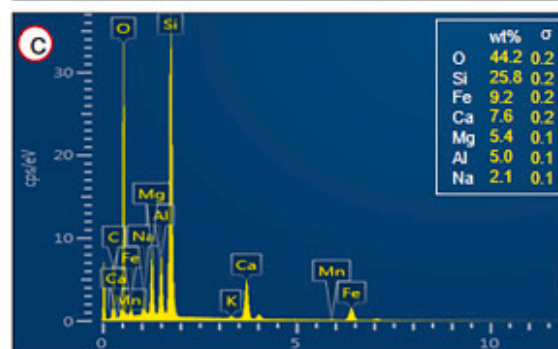
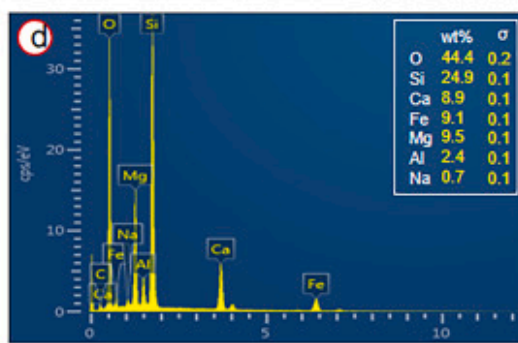
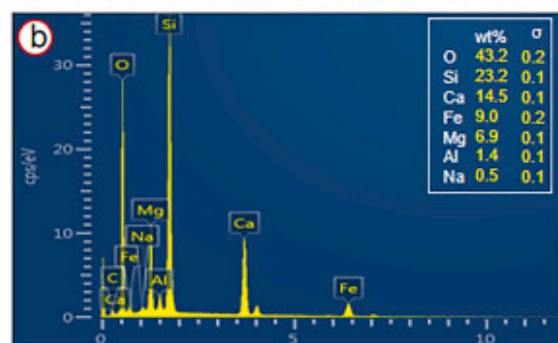
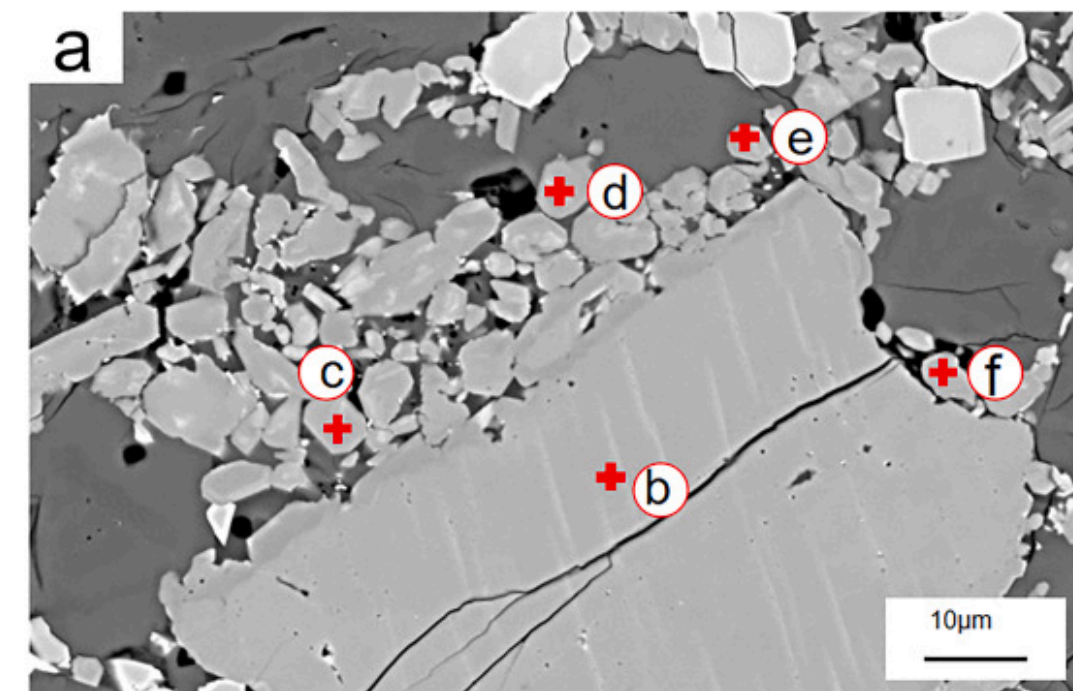
The melt percentage is lower than the CMP in this study. To quantitatively estimate the weakening effect of the melt, the stress was normalized to a strain rate of $6 \times 10^{-6} \text{ s}^{-1}$ and a temperature of 1323K. When the melt content was 2.18vol%, the normalized stress was 311.76MPa, whereas at a melt content of 5.08vol%, the normalized stress decreased to 87.14MPa. The melt weakening effect is controlled by melt topologies and melt fraction. Melt wetting grain boundaries leads to significant weakening of the aggregates even at melt content of 2–3%, which enhances shear creep by grain boundary diffusion (Tullis et al., 1996; Dimanov et al., 1998, 2000; Paterson, 2001; Zhou et al., 2012). However, if partial melt resides in triple junctions, the melt weakening increases more moderately with increasing melt content (Dimanov et al., 1998, 2000; Zhou et al., 2012). This indicates that the weakening due to melt cannot be neglected. However, further quantification requires new experiments targeted to exploration of this effect.

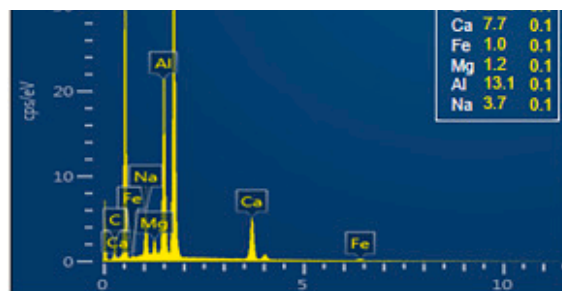
(5) Effect of reaction on rheological properties and deformation mechanisms

In our experimental study, we observed the fine-grained process of crystallization in orthopyroxene and olivine. It was observed that orthopyroxene typically forms from the core of clinopyroxene, creating a core–mantle structure with the initial clinopyroxene. Alternatively, olivine does not only nucleate in contact with cpx but also at cpx-pl as well as pl-pl grain boundaries, which may be attributed to diffusion-controlled solid–solid reactions and mass transfer processes (Fig. 6b–c). Olivine was found to be responsible for the time-dependent hardening behaviour that was seen by Zhou et al. (2017) during the creep of natural gabbro and granulite. This behaviour was brought about by the replacement of olivine for pyroxene and iron oxides. Previous experiments conducted on olivine–pyroxene aggregates at high temperatures demonstrated that the strength increases

with an increase in olivine content ([McDonnell et al., 2000](#)). The combined results of these research indicate that the interactions among pyroxene, magnetite, and olivine play a role in strengthening the lower crust's rheology. However, in our results, the newly formed olivine and pyroxene did not exhibit hardening behavior. This can be attributed to the fact that the reaction products have lower volume content (0.5vol%).

Previous studies have demonstrated that reaction products can have weakening effects ([Baïssset et al., 2024](#); [Mansard et al., 2020](#)). We observed significant grain refinement in both the “as is” samples deformed at 1423K and the water added samples deformed at 1373K. As shown in [Fig. 8](#), the number of particles (<10μm) increased, and their chemical composition differed from that of the original grains ([Fig. 16](#)). This suggests that cpx and pl continuously nucleated through mineral reactions, producing fine-grained products. The melt and reaction to form fine-grained new phases caused dissolution-precipitation creep are main process ([Herwegh et al., 2004](#); [de Ronde et al., 2005](#); [Herwegh et al., 2008](#); [Mansard et al., 2020](#); [Marti et al., 2017,2018, 2020](#); [Linckens et al., 2011](#); [Baïssset et al., 2024](#); [Nevskaya et al., 2025](#)), which induce the overall reduction in average grain size resulted in deformation under the grain-size-sensitive creep regime. Transition from dislocation creep to both dissolution-precipitation creep and grain-size-sensitive creep could lead to a notable decrease in sample strength ([Baïssset et al., 2024](#); [Bürgmann and Dresen, 2008](#); [Herwegh et al., 2011](#); [Herwegh and Berger, 2004](#); [Herwegh and Handy, 1998](#); [Herwegh and Jenni, 2001](#); [Rybacki and Dresen, 2004](#)).





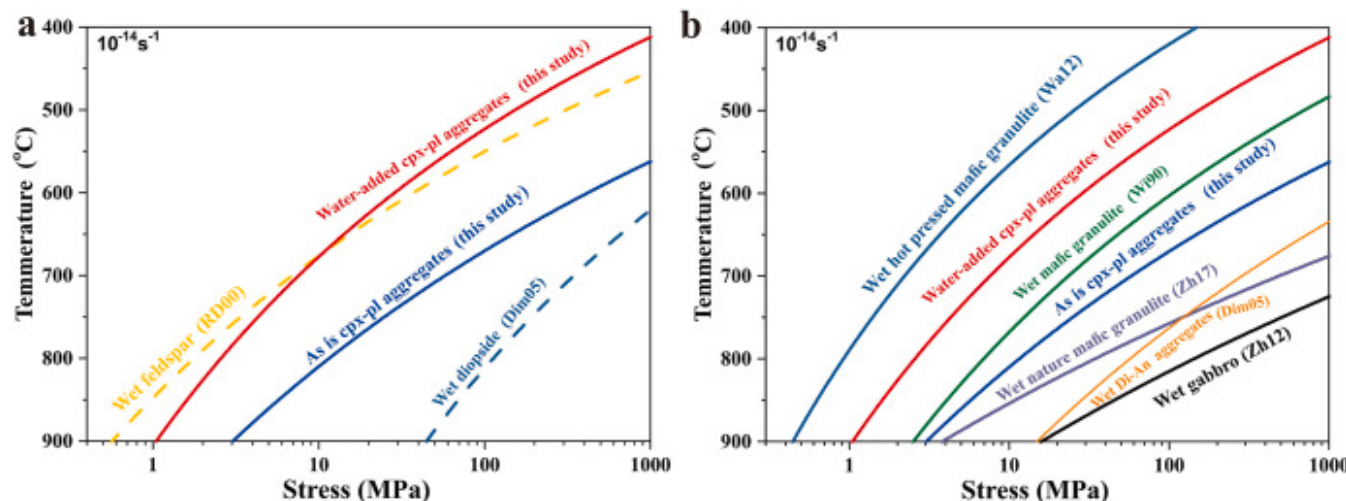
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Fig. 16. (A) SEM image of a particle refinement area in the water-added sample deformed at 1373K. The fine particles are distributed around the parent grains. (b–f) EDS spectra measured from three points (marked by red crosses) in panel (a). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4.2. Comparison with previous studies

Comparing the rheological strength of our clinopyroxene–plagioclase aggregates to that of plagioclase and clinopyroxene within the dislocation creep regime is necessary because these two minerals are the main end-member component minerals in the studied mafic granulite and are frequently thought to be representative of the rheological behavior of the CLC (Fig. 17a). The stress of our "as is" clinopyroxene–plagioclase aggregates lies between that of wet plagioclase (Rybacki and Dresen, 2000) and wet clinopyroxene (Dimanov and Dresen, 2005) under lower crustal temperature settings, based on an extrapolation to a natural strain rate of 10^{-14}s^{-1} . It is evident that our water-added samples were 3–4 times weaker than the "as is" samples, which can be attributed to the weakening effects of water and melt.



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Fig. 17. Flow strength comparison of water-added and “as is” clinopyroxene–plagioclase aggregates in this study with previous studies in the dislocation creep regime by extrapolating flow laws to a natural strain rate of 10^{-14}s^{-1} . (a) Comparing with plagioclase and clinopyroxene (b) hot-pressed granulite, natural granulite, and gabbro. Flow laws: wet anorthite aggregates (Rybacki and Dresen, 2000); wet 50% diopside and 50% anorthite aggregates (Dimanov and Dresen, 2005); wet diopside (Dimanov and Dresen, 2005); wet hot pressed mafic granulite (Wang et al., 2012); Wet mafic granulite (Wilks and Carter, 1990); wet nature mafic granulite (Zhou et al., 2017); wet gabbro (Zhou et al., 2012).

The rheological strength of our clinopyroxene–plagioclase aggregates and the granulite data from earlier investigations are contrasted in Fig. 17b. We found that our “as is” samples were >5 times weaker than the hot-pressed aggregates comprising 50% anorthite and 50% clinopyroxene (Dimanov and Dresen, 2005). The similarity in water contents (600ppm) and mineral compositions between the two experiments allows us to exclude the effects of water and mineral content. In the experiments of Dimanov and Dresen (2005), dislocation creep is major deformation mechanism. The final grain size was less than $45\mu\text{m}$, and melt was absent along the two-phase boundaries in their deformed samples. They therefore inferred that grain size reduction during initial dislocation creep led to a transition to solid state diffusion creep (ie. Coble Creep). This weakening mechanism probably also acted in our samples, but in our case the diffusion was fluid- and/or melt-assisted.

The rheological strength of the felsic granulite (Wilks and Carter, 1990) lies between that of our “as is” and water-added samples. The considerable variances in mineral composition between our samples and those reported by Wilks and Carter (1990) make it difficult to explain the similarities and variations in strength between them. Our samples are composed of 54% plagioclase and 46% clinopyroxene, whereas Wilks and Carter's samples are composed of 58% amphibole, 40% plagioclase, 1% clinopyroxene, and 1% orthopyroxene. Additionally, the water content in the minerals analyzed by Wilks and Carter (1990) was not specified, which makes it impossible to effectively discuss this aspect.

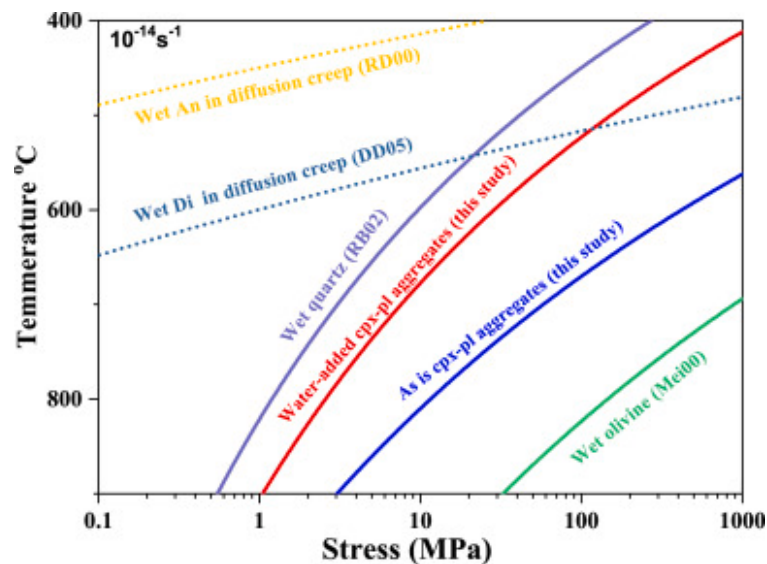
Compared to the mafic granulite studied by Wang et al. (2012), which is composed of 57% plagioclase, 14% orthopyroxene, 24% clinopyroxene, and 5% opaque minerals, our “as is” and water-added samples demonstrate significantly higher strengths 9–10 times and 2–3 times, respectively (Fig. 17b). These disparities can be attributed to two primary factors. First, the water content in Wang et al.'s samples (1600–2800ppm H₂O) is considerably higher than that in our clinopyroxene–plagioclase aggregates. The more pronounced weakening effect of water at higher concentrations likely contributes to the substantial disparity in strength. Second, cleavage-induced fractures during hot pressing caused an anisometric shape of clinopyroxene grains, resulting in weak SPO (Fig. 11) and CPO fabrics (Fig. 12). Thus, mechanical anisotropy was added, improving the strength of our clinopyroxene–plagioclase aggregates (Hansen et al., 2012a, Hansen et al., 2012b; Ji et al., 2000).

Compared to the natural mafic granulite studied by Zhou et al. (2017), our “as is” clinopyroxene–plagioclase aggregates display lower strength. Given that the water content remains consistent across the studies, this disparity in strength can be attributed to several factors. According to extrapolations made using previously published dislocation creep rheology laws for both minerals under wet settings (Bystricky et al., 2016; Chen et al., 2006), orthopyroxene is only slightly stronger than clinopyroxene. The heightened presence of orthopyroxene in the natural mafic granulite likely contributes to its overall strength. Furthermore, the observed variance in stress levels may partially stem from differences in grain size or disparities in the dominant deformation mechanisms between various experimental studies.

4.3. Geological implications

Extensive observations have indicated that the lower continental crust in the granulite facies has the ability to retain significant amounts of water that is structurally bound within minerals that are considered to be anhydrous (Johnson, 2006; Xia et al., 2006; Yang et al., 2008). Compared to wet quartz aggregates, which depict the upper crust at comparable pressure and temperature levels, our clinopyroxene–plagioclase aggregates exhibit notably greater strength when extrapolated to a natural strain rate of

10^{-14} s^{-1} (Fig. 18). Still, because of lower temperatures and the predominance of brittle deformation features, it is commonly accepted that the upper crust is stronger than the lower crust (Bürgmann and Dresen, 2008; Kohlstedt et al., 1995; Ranalli and Murphy, 1987). Our clinopyroxene–plagioclase aggregates are much weaker than wet olivine aggregates, which are indicative of the upper mantle (Mei et al., 2000). Our experimental findings illustrate that clinopyroxene–plagioclase aggregates containing 2000ppm of water are more than ten times weaker than the wet upper mantle. Even when the water content in clinopyroxene–plagioclase aggregates is reduced to 600ppm, the contrasting rheological properties between the weaker lower continental crust and the stronger underlying upper mantle remain evident. According to these findings, the strength profile of the continental lithosphere is consistent with the 'jelly sandwich' model proposed by W. P. Chen and Molnar (1983) since the lower crust is weaker than the upper crust and upper mantle. Furthermore, this model is reinforced by numerous global geodetic and geological observations (Bürgmann and Dresen, 2008; Homburg et al., 2010).



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Fig. 18. Comparisons of flow strength between our “as is” and water-added clinopyroxene–plagioclase aggregates with those of quartz and olivine aggregates in the dislocation creep regime, as well as anorthite and diopside aggregates in the diffusion creep regime, by extrapolating flow laws to a natural strain rate of 10^{-14} s^{-1} . wet quartz aggregates (Rutter and Brodie, 1988); wet

olivine ([Mei and Kohlstedt, 2000](#)); wet anorthite aggregates ([Rybacki and Dresen, 2000](#)); wet diopside ([Dimanov and Dresen, 2005](#)).

The creep strength of wet granulite is significantly lower than that of an olivine aggregate, which may result in delamination between the granulite-bearing lower crust and eclogite-bearing lower crust or upper mantle, particularly if the olivine phase in the upper mantle is gravitationally unstable in relation to the asthenosphere. The weakened CLC is a crucial factor in the delamination and submergence of eclogite roots into the deeper mantle ([Meissner and Mooney, 1998](#)). Moreover, the presence of a weak lower crust hampers the effective transmission of shear stress generated by mantle convection to the upper crust, leading to mechanical decoupling between the upper crust and upper mantle ([Royden, 1996](#)). This decoupling promotes the flow of the weak lower crust, enabling it to absorb lateral pressure gradients induced by surface topography ([Shen et al., 2001](#)). Within the lower crust, the flow tends to align anisotropic minerals, such as pyroxene and plagioclase, in a specific direction, which explains the observed seismic anisotropy and layered reflectors in the CLC ([Ross et al., 2004](#); [Meissner et al., 2006](#); [Moschetti et al., 2010](#)). The delamination and submergence of eclogite roots into the deeper mantle, driven by the weakness of the CLC, contribute to lithospheric thinning and asthenospheric upwelling, further facilitating the flow of the lower crust. During dislocation processes, neocrystallization reduced grain size assisting a transition to solution- and melt-accommodated diffusion creep. Thus, the proportion of diffusion creep and dissolution-precipitation creep gradually increases with strain increasing, which induce to decrease flow stress ([Fig. 18](#)). Deformation of the lower crust, originally controlled by dislocation creep, transitions to being dominated by diffusion creep, resulting in a decrease in rheological strength ([Fig. 18](#)). In this process, localization in the lower crust is characterized by fine-grained diffusion creep with strain increasing, which caused to the formation of ductile shear zones ([Rybacki et al., 2008, 2009](#); [Wehrens et al., 2016](#); [Nevskaya et al., 2025](#)).

5. Conclusions

Hot-pressed samples of “as is” and water-added clinopyroxene–plagioclase aggregates were subjected to deformation using a Paterson-type gas-medium deformation apparatus. The deformation was carried out under a confining pressure of 300MPa, at a temperature of 1173–1423K, and strain rates of 6×10^{-6} to $1 \times 10^{-5} \text{ s}^{-1}$. Consequently, axial stresses of 15–464MPa were induced, resulting in a total strain of 14%–20%.

1. We obtained the flow law for “as is” clinopyroxene–plagioclase aggregates (~600ppm)

$\dot{\epsilon} = 10^{6.7 \pm 0.7} \sigma^{3.6 \pm 0.6} \exp \left(-\frac{504 \pm 85 \text{ KJ/mol}}{RT} \right)$. For water-added clinopyroxene–plagioclase aggregates (~2000ppm), the flow law is $\dot{\epsilon} = 10^{2.3 \pm 0.4} \sigma^{3.7 \pm 1.4} \exp \left(-\frac{347 \pm 48 \text{ KJ/mol}}{RT} \right)$.

2. Microstructural observations revealed that the deformed samples exhibited a shape-preferred orientation (SPO) and crystallographic preferred orientation (CPO) with subgrain boundaries in coarse-grained domains (constituting ~80vol%), indicating that the primary deformation mechanism is dislocation creep. Additionally, the presence of melt, neoformed phases (olivine+orthopyroxene) and abundant fine-grained particles (<10μm) all show subsidiary dissolution-precipitation creep, and the subsidiary dissolution-precipitation creep played an important role under higher temperature and higher water content conditions. The overall reduction in average grain size resulted in deformation under the grain-size-sensitive creep regime, leading to a notable decrease in sample strength.
3. The data obtained support the notion that the lower crust is comparatively weaker than the upper crust and uppermost mantle. This observation aligns with the conceptual “jelly sandwich” model of the continental lithosphere.

CRedit authorship contribution statement

Qianqian Zhang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yongsheng Zhou:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Jianfeng Li:** Methodology, Formal analysis. **Tongbin Shao:** Writing – review & editing. **Maoshuang Song:** Methodology.

Data availability

The experiment data files used in this study are available through <https://data.mendeley.com/preview/mrwyznd6kc?a=18575e1d-1ca2-4f69-9042-e8d90103b294>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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